

NutriMatch: Web-Based Clinical Nutrition Consultation Management System



**A Capstone Project Proposal
Presented to the Faculty of the
Information and Communications Technology Program
STI College Davao**

**In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Technology**

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May 2026

ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH: NutriMatch: Web-Based Clinical Nutrition Consultation Management System

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This capstone project proposal titled **NutriMatch: Web-Based Clinical Nutrition Consultation Management System**, prepared and submitted by **King Paolo L. Chiu, Ivy Hope O. Alba, Julia Niel B. Bulalague and Kent Daryll V. Leabres**, in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, has been examined and is recommended for acceptance and approval.

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INTRODUCTION

Project Context

Nutrition serves as both a frontline defense against chronic disease and a therapeutic instrument for managing conditions that impose devastating consequences on individuals and healthcare systems. Globally, noncommunicable diseases (NCDs) account for approximately 74 percent of all deaths worldwide annually, with cardiovascular disease, type 2 diabetes, hypertension, obesity, and chronic kidney disease representing the majority of this preventable mortality; unhealthy diet is among the four principal modifiable risk factors driving this burden, alongside tobacco use, physical inactivity, and the harmful use of alcohol (World Health Organization, 2023), prompting the WHO to explicitly recommend the integration of structured Medical Nutrition Therapy (MNT) into primary healthcare systems as a cost-effective, evidence-based strategy. Medical Nutrition Therapy, defined by the Academy of Nutrition and Dietetics (2024) as a systematic clinical approach delivered through the four-phase Nutrition Care Process (NCP): nutritional assessment, nutrition diagnosis, nutrition intervention, and nutrition monitoring and evaluation, has been shown to produce measurably superior clinical outcomes compared to unstructured dietary advice when supported by digital infrastructure (Evert et al., 2019; Kelly et al., 2016).

In the Philippines, the global NCD burden is intensified by a dual burden of malnutrition, with the Food and Nutrition Research Institute (2024) reporting persistent stunting and wasting among children alongside accelerating overweight and obesity in adults, while hypertension affects nearly one in three Filipino adults and type 2 diabetes prevalence continues rising across all income groups. Despite this growing burden, the tools available to Registered Nutritionist-Dietitians (RNDs) remain largely manual, fragmented, and disconnected from modern digital health infrastructure, many RNDs still rely on paper-based documentation, printed FNRI Food Exchange Lists, spreadsheets, and informal messaging applications, generating inefficiencies, clinical risks, and non-compliance with the Data Privacy Act of 2012 (RA 10173). While the COVID-19 pandemic accelerated telehealth adoption, most platforms in use are general-purpose tools designed for physicians rather than dietitians, and international systems such as Nutrium, Healthie, and Practice Better are

calibrated for Western dietary patterns and do not incorporate the FNRI Food Exchange Lists, offer native client-RND matching, or address RA 10173 compliance, gaps directly confirmed through the proponents' needs assessment, which documented the absence of any platform simultaneously addressing Philippine food data, NCP workflow support, and RA 10173 data privacy requirements.

These systemic challenges are acutely concentrated in Davao City, the sole highly urbanized city and regional center of Region XI, with a population of 1,848,947 as of July 1, 2024 (PSA, 2025). Data from the Department of Health Davao Region (2025) identified hypertensive disease as the second most prevalent illness in the region with 36,856 reported cases, while diabetes mellitus ranked among the leading causes of mortality, figures consistent with national trends documented by the FNRI (2024) and confirming that the lifestyle disease burden in Davao City and Region XI is among the most pressing in Mindanao. The clinical nutrition workforce in Davao City operates entirely on paper-based records, handwritten dietary assessments, and manual nutrient computation, a fragmented practice environment firsthand confirmed through semi-structured interviews with licensed RNDs Prince Immanuel L. Chiu and Joerine Trexie Jimenez, and a structured questionnaire administered to Joerine Trexie Jimenez and Gyle Francis Forones, during the proponents' needs assessment, documenting the absence of any locally designed, clinically structured, or RA 10173-compliant digital platform in Davao City's hospital dietetics departments. NutriMatch is developed as a direct, locally grounded response to this documented gap, designed on an RND-first model in which individual licensed practitioners deploy the system independently in their own clinical practice, with formal endorsement from hospital dietetics departments providing institutional legitimacy and clinical context.

Purpose and Description of the Project

NutriMatch modernizes Medical Nutrition Therapy delivery by replacing manual, fragmented workflows with a unified digital platform. Built on Laravel, Nuxt.js, and MySQL, it provides a secure, role-based environment with RESTful APIs supporting three user roles: Registered Nutritionist-Dietitians, clients, and administrators.

The system opens with a secure registration and professional verification module that validates RND credentials, ensuring only licensed practitioners access clinical features. A client-RND matching feature allows clients to search for dietitians by specialty area, language preference (Tagalog, Cebuano, Ilocano, English), and consultation type (video, chat, in-person), with profiles displaying reviews and ratings to support informed decision-making. Authenticated users are directed to role-calibrated dashboards: RND dashboards present the patient panel, upcoming appointments, key performance indicators, and automated consultation summaries; client dashboards display health journey progress and appointment summaries; and administrator dashboards provide centralized oversight of dietitian verification status, commission tracking, activity logs, and system audit trails.

At the clinical core lies a digital Nutrition Care Process module operationalizing all four MNT phases. The assessment phase automates computation of BMI, BMR, and nutritional risk classification using adapted NRS-2002 scoring with WHO Asia-Pacific BMI thresholds. The diagnosis phase enables structured PES statement documentation. The intervention phase provides guided care planning incorporating calorie counts, 24-hour dietary recalls, food diaries, food frequency questionnaires, and Individual Dietary Diversity Score assessments. The monitoring and evaluation phase integrates clinical charting with automated alerts and longitudinal outcome visualization. Supporting the intervention phase, the disease-specific meal planning module generates condition-specific diet templates for diabetes mellitus, hypertension, renal disease, and weight management, drawing from both the FNRI Food Exchange Lists (4th Edition, 2020) and USDA FoodData Central, a dual-source approach that ensures meal plans reflect actual Filipino dietary patterns while

maintaining globally validated nutrient accuracy.

NutriMatch includes an automated reporting and progress monitoring module serving both RNDs and clients through interactive dashboards that visualize weight, BMI, blood pressure, blood glucose, lipid profiles, and dietary adherence, generating consultation summaries and progress reports after each session. RNDs are further supported by a digital resource repository for uploading nutrition education materials and an automated reminder module delivering notifications for appointments, meal logging, and motivational prompts. Billing and payment are managed across dedicated RND, client, and administrative modules, with all system actions captured through automated administrative documentation. The platform complies with RA 10173 (Data Privacy Act of 2012) and aligns with RA 10918 (Philippine Nutrition Act), implementing SSL/TLS encryption, hashed credential storage, role-based access controls, audit logging, and digital informed consent collection.

NutriMatch operates under several constraints. Its full MNT feature set requires an established RND-client relationship and a stable internet connection; no offline mode is supported. Nutritional calculations are bounded by the FNRI Food Exchange Lists (4th Edition, 2020) and USDA FoodData Central, and automated outputs are intended to assist, not replace, clinical judgment. Video conferencing, payments, and SMS notifications rely on third-party APIs outside NutriMatch's direct control. PRC license verification is manual, and real-time detection of post-verification changes cannot be guaranteed. Deployment compliance depends on the hosting institution's server configuration. As a capstone system, it does not include formal NPC Data Processing System registration, deploying organizations assume that responsibility. The system also lacks an automated data retention schedule and granular consent revocation, and communications are HTTPS-secured at the transport layer but not end-to-end encrypted.

Objectives of the Study

General Objective:

To develop a web-based clinical nutrition consultation management system for Registered Nutritionist-Dietitians to efficiently handle medical nutrition therapy information of their clients.

Specific Objectives:

- To implement an RND-client matching system filtered by specialty, language, and consultation modality, including video conferencing, with profiles featuring ratings and peer reviews.
- To develop a digital Nutrition Care Process integrating automated nutritional risk assessment, structured Medical Nutrition Therapy documentation, patient charting, and recording of laboratory data for clinical management.
- To incorporate disease-specific and localized meal planning modules using Philippine food composition data and international nutrient databases to generate accurate diet templates for chronic conditions.
- To establish a digital repository for nutrition and physical activity resources, integrated with an automated reminder system, to support client self-management and behavioral adherence between sessions.
- To construct a secure billing module for managing consultation fees and transactions on both client and dietitian sides, coupled with an administrative system for commission tracking and platform oversight.

Scope and Limitations of the Study

NutriMatch is a web-based clinical nutrition consultation management system accessible through any standard internet browser. Three user roles govern all interactions: Registered Nutritionist-Dietitians, clients, and administrators, each within a role-based access control structure.

The system encompasses eight functional modules: Client-RND Matching, Pre-Consultation Nutritional Screening, Digital Nutrition Care Process, Disease-Specific Meal Planning, Automated Reporting and Progress Monitoring, Digital Resource Repository, Automated Reminder and Habit-Support, and Billing and Payment Management, with Administrative Dashboard and Documentation modules providing platform-level oversight. All components are implemented in compliance with Republic Act No. 10173 and in alignment with Republic Act No. 10918.

NutriMatch operates within several defined constraints. Clinical features are accessible only within a formal RND-client relationship, and a stable internet connection is required with no offline mode available. Nutritional calculations are constrained by the FNRI Food Exchange Lists (4th Edition, 2020) and USDA FoodData Central, and automated computations are intended to support, not replace, RND professional judgment. Video conferencing, payment processing, and SMS notifications are delivered through third-party APIs governed by their respective providers.

PRC license verification is conducted through manual administrator cross-referencing against the PRC online verification portal rather than an automated government API. As a capstone system, NutriMatch does not include formal Data Processing System registration with the National Privacy Commission; organizations deploying it for clinical use are responsible for this independently. Communications are protected at the transport layer through HTTPS but are not end-to-end encrypted at the application layer.

CHAPTER II

Review of Related Literature

This chapter reviews the literature and systems relevant to NutriMatch into two sections: a Review of Related Literature examining the clinical frameworks, empirical evidence, and regulatory mandates establishing the problem NutriMatch addresses; and a Review of Related Systems analyzing existing platforms to identify gaps NutriMatch is designed to fill.

Medical Nutrition Therapy (MNT) is a evidence-based clinical approach to chronic and diet-related conditions, defined by the Academy of Nutrition and Dietetics (2024) through the four-phase Nutrition Care Process (NCP) framework, which standardizes clinical documentation and outcome tracking. Evert et al. (2019) demonstrated that structured, dietitian-led MNT achieved significantly better glycemic control, lipid profiles, and blood pressure outcomes compared to standard care.

The WHO (2023, 2024) endorses dietary counseling as a best-buy intervention for noncommunicable disease prevention, while the FNRI (2024) documented persistent malnutrition among children alongside accelerating overweight and obesity in adults. The proponents' needs assessment confirmed that digital tool adoption among RNDs in Davao City remains low due to the absence of platforms addressing Philippine food data, NCP workflow support, and RA 10173 compliance.

Kelly et al. (2016) and Barnett et al. (2023) confirm that effective digital dietary interventions require structured dietitian-led delivery, follow-up protocols, and behavioral support. Banal et al. (2024) demonstrated that MyFitnessPal produced significant nutrient estimation errors when applied to Filipino adults with obesity, reinforcing the need for locally calibrated food composition data and WHO Asia-Pacific BMI standards in NutriMatch's assessment algorithms

Health information systems in the Philippines must comply with Republic Act No. 10173 and NPC Circular No. 16-01, mandating role-based access, audit logging, and informed consent. Poulsen et al. (2024) found that many regional health apps lack these protections, gaps NutriMatch addresses.

Review of Related System

Nutrium (2024) is a cloud-based platform used by dietitians in Europe and North America offering meal planning, client management, and progress tracking. Its food database contains no Filipino food entries, lacks NCP documentation support including PES statement templates and monitoring frameworks, and does not offer pre-consultation risk screening, WHO Asia-Pacific BMI computation, client-RND matching, or RA 10173-compliant billing.

MyFitnessPal (2024) is a consumer wellness application featuring a crowdsourced food database, barcode scanning, and macronutrient tracking. It lacks role-based access, structured clinical documentation, and RA 10173 compliance. Banal et al. (2024) found it produced significant discrepancies in energy and macronutrient estimates when applied to Filipino adults with obesity, making it unsuitable for RNDs managing renal or diabetic diets where nutrient precision determines clinical safety.

Dela Cruz et al. (2023) developed NutriCare PH, a web-based capstone system at Mapúa University for tracking pediatric nutritional status using WHO child growth standards. Its scope excluded clinical workflows for licensed RNDs managing adult chronic conditions, with no MNT documentation, PES statement generation, appointment scheduling, or client-RND matching implemented, and without full RA 10173 compliance mechanisms.

Table 1: Comparison Matrix of Related Systems

Features / Criteria	Nutrium (2024)	MyFitnessPal (2024)	NutriCare PH (Dela Cruz et al., 2023)	NutriMatch (Proposed System)
Food Database & Nutritional Computation				
Validated food database	✓	~	✗	✓
Philippine food data (FNRI Food Exchange Lists)	✗	✗	✗	✓
USDA FoodData Central integration	✓	✗	✗	✓
Automated BMI / BMR / TDEE computation	✗	✓	~	✓
Disease-specific diet templates (DM, HTN, Renal)	~	✗	✗	✓
Clinical Nutrition Care Process (NCP)				

Full four-phase NCP documentation	✗	✗	✗	✓
NRS-2002 nutritional risk screening	✗	✗	✗	✓
Structured PES statement documentation	✗	✗	✗	✓
Laboratory data recording	✗	✗	✗	✓
Clinical outcome report generation	✗	✗	✗	✓
Client–RND Matching & Consultation Management				
Client–RND matching (specialty, language, modality)	✗	✗	✗	✓
RND ratings and peer reviews	✗	✗	✗	✓
Appointment scheduling and management	✓	✗	✗	✓
Video / chat / in-person consultation support	~	✗	✗	✓
Task assignment and progress tracking	✗	✗	✗	✓
Progress Monitoring & Resource Support				
Client progress / self-monitoring dashboard	✓	✓	~	✓
RND patient panel and KPI overview	✓	✗	✗	✓
Nutrition resource repository (RND-managed)	~	✗	✗	✓
Automated reminders and habit-support prompts	✗	✗	✗	✓
Billing & Financial Management				
Consultation fee billing module	✓	✗	✗	✓
Integrated payment gateway	✗	✗	✗	✓
Platform commission and earnings tracking	✗	✗	✗	✓
Data Privacy, Security & Compliance				
RA 10173 (Data Privacy Act) compliance	✗	✗	~	✓
Audit logging of clinical/admin actions	✗	✗	✗	✓
PRC license credential verification	✗	✗	✗	✓
RA 10173 informed consent management	✗	✗	✗	✓

Synthesis

The reviewed literature and systems confirm that structured, digitally supported MNT improves patient outcomes, and that the Philippine dietetic profession faces a documented, unmet need for a locally grounded digital platform. The Comparison Matrix in Table 1 confirms that Nutrium, MyFitnessPal, and Dela Cruz et al. (2023) each score absent or partial across the six functional dimensions evaluated, gaps that directly inform the design of NutriMatch: its dual FNRI and USDA FoodData Central database architecture, fully embedded four-phase NCP with structured PES documentation and automated risk assessment, WHO Asia-Pacific BMI standards, role-based access control with professional credential verification, and RA 10173-compliant data handling with SSL/TLS encryption, audit logging, and informed consent mechanisms.

Methodology

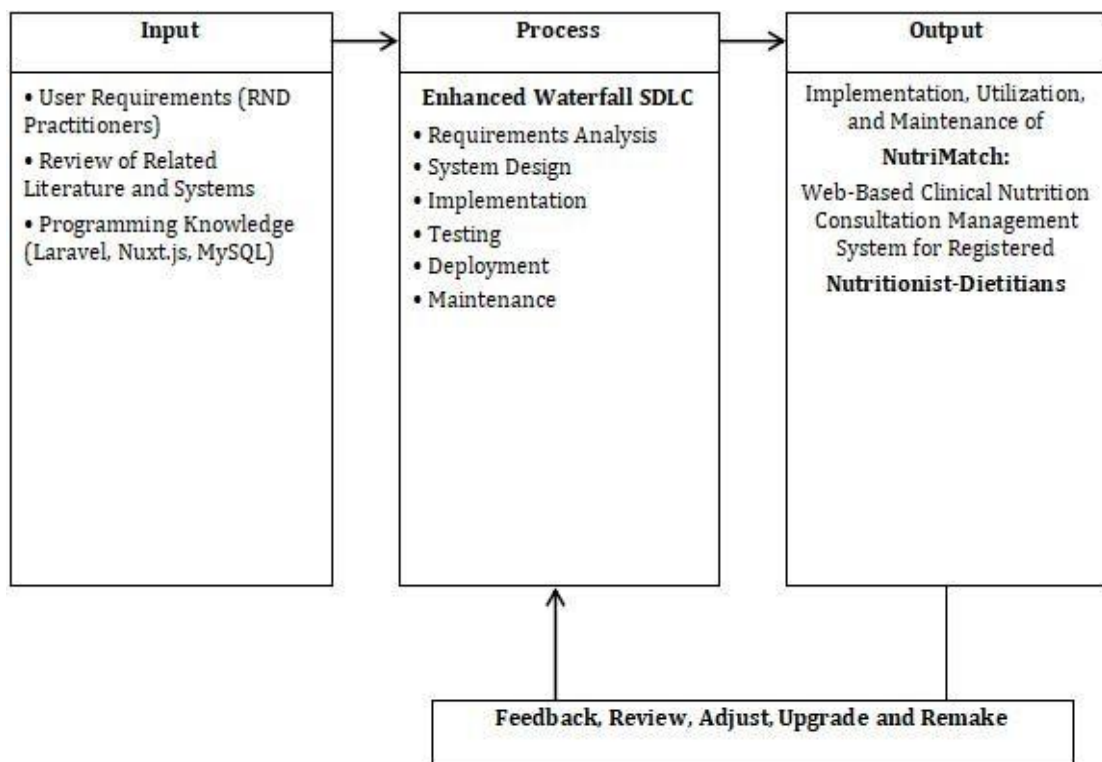
This chapter presents the methodology governing the development of NutriMatch, organized around an Input-Process-Output Conceptual Framework and the Enhanced Waterfall SDLC with Feedback. Inputs include user requirements from licensed RND practitioners, the Review of Related Literature and Systems, the FNRI Food Exchange Lists, clinical nutrition standards, regulatory requirements under RA 10173 and RA 10918, and the programming knowledge base covering Laravel, Nuxt.js, and MySQL. These feed into the seven-phase Enhanced Waterfall SDLC, producing the implementation, utilization, and maintenance of NutriMatch as its output.

The proponents adopted the Enhanced Waterfall model with Feedback, a structured and sequential methodology preserving the phase-gate discipline of classical Waterfall while incorporating four controlled mechanisms at each phase boundary. Each phase must be substantially completed before the next begins, with all decisions traceable to a formally documented requirements baseline. At the close of each phase, a formal Review is conducted against functional and nonfunctional requirements and RA 10173 compliance mandates. Where discrepancies are identified, a Feedback loop is activated, with an Adjust response for targeted modifications or an Upgrade/Remake response when the identified gap is systemic.

The Requirements Determination Phase gathered requirements through literature review, semi-structured interviews, and structured questionnaires with licensed RNDs. The Requirements Analysis Phase modeled validated requirements using Use Case Diagrams, BPMN and Activity Diagrams, Domain Class Diagrams, and Sequence Diagrams, culminating in a Software Requirements Specification. The System Design Phase translated these into a high-level architecture covering the MySQL database schema, RESTful API specifications, and Nuxt.js frontend layouts. The Implementation Phase coded all eight modules under strict coding standards and Git version control. The Testing Phase executed unit, functional, integration, and User Acceptance Testing, with every defect initiating a Feedback loop. The Deployment Phase will transition the system to a cloud hosting platform with SSL/TLS configuration and role-specific user manuals. The Maintenance Phase will provide post-deployment support through structured monitoring and issue triage.

The Conceptual Framework in Figure 1 makes every design decision traceable to a documented source. The Input layer reflects Philippine dietetic practice realities and regulatory mandates. The Process layer formalizes how inputs become a deliverable system through seven sequential SDLC phases. The Output layer defines success as a fully implemented, utilized, and maintained system, with the closed-loop architecture ensuring the Maintenance phase feeds findings back into the process.

Figure 1. Conceptual Framework



Technical Background

• Technologies to be Used

The technologies to be used in the proposed system are presented through a Technology Viewpoint diagram, which illustrates the overall client-server architecture of NutriMatch. The diagram shows how the Nuxt.js front-end, Laravel API back-end, and MySQL database interact across three principal nodes: the Application Node, the Database Node, and the External Services Node, to support centralized clinical nutrition consultation operations. It also demonstrates how users can access the system through desktop, mobile, and tablet devices via an internet connection, ensuring accessibility, secure communication, and efficient data management, in full compliance with RA 10173 data minimization requirements.

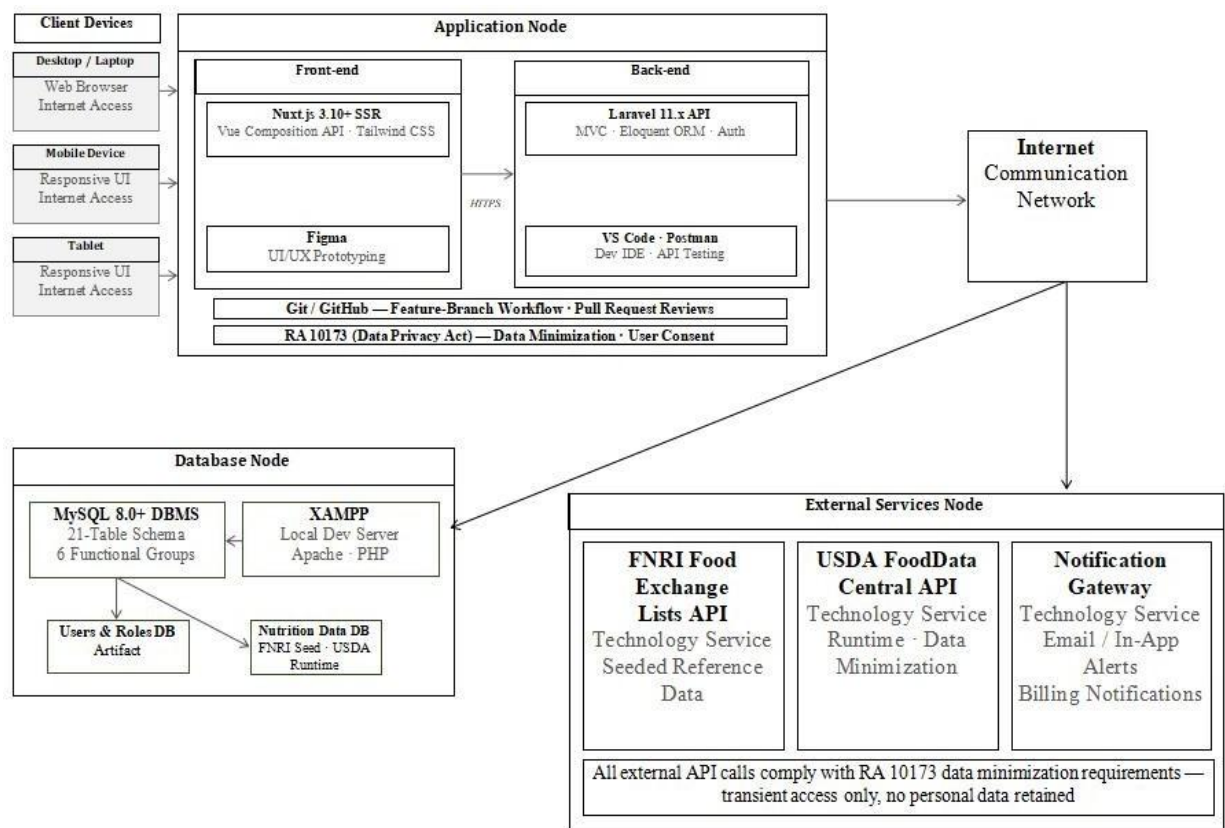


Figure 2. Technology Viewpoint Diagram of NutriMatch

As shown in *Figure 2*, NutriMatch adopts a three-node architecture. The Application Node houses the Nuxt.js 3.10+ front-end utilizing the Vue Composition API and Tailwind CSS and the Laravel 11.x back-end, which implements the MVC pattern with Eloquent ORM and built-in authentication scaffolding. The Database Node employs MySQL 8.0+ to manage the twenty-one-table schema across six functional groups, with FNRI Food Exchange Lists integrated as seed data. The External Services Node connects at runtime to the USDA FoodData Central API and a notification gateway for email and in-app alerts. All components are version-controlled via Git and GitHub using a feature-branch workflow, developed using Visual Studio Code 1.85+ and tested with Postman 11.x, with UI/UX prototyped in Figma.

- **Calendar of Activities**

The detailed sequence of activities that the proponents underwent in completing the project, covering all seven phases of the Enhanced Waterfall SDLC with Feedback, including the activities, purpose, persons involved, and resources needed in chronological order of execution, is presented in *Figure 3* of the Appendices.

- **Resources**

The specific hardware and software resources that the proponents used in completing the project are enumerated in *Figure 4* and *Figure 5* of the Appendices.

Requirement Determination

NutriMatch requirements were determined through semi-structured interviews and a structured questionnaire administered to licensed Registered Nutritionist-Dietitians in Davao City, supported by reviews of Medical Nutrition Therapy literature, Philippine regulations, and existing nutrition platforms. The findings were compiled into a Software Requirements Specification document that guided the system's design and development.

To gather professional insight, the proponents conducted semi-structured interviews with two licensed RNDs, Prince Immanuel L. Chiu and Joerine Trexie Jimenez. The interviews explored their clinical workflows, documentation practices, challenges with existing tools, and preferred features for a digital nutrition consultation platform. The open-ended format allowed respondents to provide detailed insights beyond what a structured questionnaire alone could capture.

The proponents administered a structured questionnaire as shown in *Figure 6* in the Appendices, to two hospital-based RNDs in Davao City, Joerine Trexie Jimenez and Gyle Francis Forones, covering workflow, challenges, feature importance, and consultation needs. Both relied on manual computations, FNRI Food Exchange Lists, spreadsheets, and messaging applications, with meal planning and patient monitoring identified as the most time-consuming tasks. Recurring challenges included difficulty tracking patient progress and patient miscommunication. Both respondents rated nearly all proposed NutriMatch features as Very Important and confirmed willingness to use the system if developed. These findings provided a validated basis for the functional and non-functional requirements documented in the Software Requirements Specification.

Requirement Analysis

This section presents the requirements analysis of NutriMatch through four modeling dimensions: the Functional Model, Process Model, Data Model, and Behavioral Model, providing a complete specification framework aligned with the validated requirements from the Requirements Determination phase.

The Functional Model describes the system's functions, user interactions, and access boundaries through Use Case Diagrams, identifying the three primary actors, the Client, the RND, and the Administrator, and defining their accessible functions as a traceable reference for translating interview and questionnaire findings into NutriMatch's eight core modules.

The Process Model describes the sequence of activities and workflows through which the system carries out its clinical, administrative, and transactional processes using BPMN Diagrams for high-level workflows, task handoffs, and decision points among actors and external systems, and Activity Diagrams for detailed conditional logic within each module, including automated computations such as NRS-2002 risk scoring and TDEE calculations.

The Data Model describes the system's persistent data entities, their attributes, and inter-entity relationships using Domain Class Diagrams, distinguishing between data permanently stored in the MySQL database and external nutrient data temporarily accessed from the FNRI FCT API and USDA FoodData Central API during runtime without being persisted, ensuring data minimization compliance with RA 10173 is architecturally explicit.

The Behavioral Model describes how the system responds to events through the chronological interaction of its components using Sequence Diagrams, which emphasize inter-component communication across processes such as nutritional risk assessment, meal plan generation, and consultation payment, making them suitable for specifying the implementation logic of NutriMatch's Laravel-based RESTful API architecture.

Requirement Documentation

This document presents the formal Software Requirements Specification (SRS) for NutriMatch, providing the authoritative definition of the system's functional and non-functional requirements to guide all subsequent phases of the software development lifecycle.

Functional Model

The Functional Model, illustrated through Use Case Diagrams in *Figures 7 through 12*, defines the three primary actors, the Client, the RND, and the Administrator, and the functions each may perform. The RND-Client Matching Module (*Figure 7*) allows Clients to search and filter RNDs by specialty, language, and consultation modality, with the Administrator verifying RND credentials. The Pre-Consultation Screening Module (*Figure 8*) allows Clients to enter biometric data, upon which the system automatically computes BMI and BMR, classifies nutritional risk, and grants both parties access to the screening summary. The Digital NCP Module (*Figure 9*) allows the RND to conduct NRS-2002-based nutritional risk assessment, document MNT and NCP records, draft PES Statements, and generate Clinical Outcome Reports. The Meal Planning Module (*Figure 10*) enables the RND to select disease-specific diet templates for Diabetes, Hypertension, Renal disease, and Weight Management, querying the FNRI Food Exchange Lists and USDA FoodData Central. The Digital Repository Module (*Figure 11*) supports RND upload of nutrition education materials and configuration of automated reminders. The Billing Module (*Figure 12*) allows Clients to manage payments, Dietitians to track earnings, and Administrators to manage commission structures and transaction logs. Non-functional requirements mandate role-based access control, three-second page load, encryption of all personal and clinical data, and RA 10173 compliance.

Process Model

The Process Model employs four BPMN Diagrams (*Figures 13 through 16*). *Figure 13* models RND profile setup, client filtering, and match request dispatch. *Figure 14* models automated BMI, BMR, and NRS-2002 computation, dietary history collection, PES statement encoding, and disease-specific meal plan generation from FNRI and USDA data. *Figure 15* models automated reminders, client progress

logging, RND plan adjustment, and outcome report generation. *Figure 16* models consultation fee input, invoice generation, payment gateway processing, commission calculation, RA 10173 audit log generation, and digital receipt issuance.

The Activity Diagrams (*Figures 17 through 22*) provide module-level conditional logic. *Figure 17* shows consultation request dispatch and clinical module unlock upon acceptance. *Figure 18* shows the Calculation Engine generating BMI, BMR, TDEE, and NRS simultaneously for both the Client and the RND's NCP Assessment interface. *Figure 19* shows sequential draft-save across all four NCP phases and consultation summary publication upon finalization. *Figures 20 through 22* model FNRI-FCT API lookup with USDA fallback and meal plan generation, resource assignment and reminder dispatch, and invoice creation with payment gateway processing and financial report generation respectively.

Data Model

The Data Model is represented through five Domain Class Diagrams (*Figures 23 through 27*). The `rnd_client_relationships` entity (*Figure 23*) anchors all downstream clinical records. The `ncp_records` entity (*Figure 24*) encapsulates all four NCP phases in a single longitudinal record. The meal planning hierarchy (*Figure 25*), composed of `meal_plans`, `meal_plan_meals`, and `meal_plan_food_items`, persists only a plain-text `food_name` for externally sourced items with no nutrient payload written to the database, maintaining RA 10173 data minimization compliance. The `invoices` entity (*Figure 26*) freezes commission amounts at creation. *Figure 27* introduces the `audit_logs` entity recording an immutable trail of all significant actions per the Data Privacy Act of 2012.

Behavioral Model

The Behavioral Model is represented through six Sequence Diagrams (*Figures 28 through 33*). *Figure 28* models client-initiated RND search and `rnd_client_relationships` activation with audit logging. *Figure 29* models Mifflin-St Jeor BMR computation, TDEE derivation, WHO Asia-Pacific BMI classification, and NRS-2002 scoring, persisted and rendered for both the Client and the RND. *Figure 30* models the Digital Nutrition Care Process sequence, capturing the four-phase NCP clinical documentation flow with draft-save behavior and consultation summary

publication upon finalization. *Figure 31* models the Disease-Specific Meal Planning sequence, illustrating FNRI FCT and USDA FoodData Central API lookup, nutrient aggregation, and finalized meal plan delivery to the Client. *Figures 32 and 33* model resource upload and reminder scheduling, and end-to-end payment authorization with RA 10173 audit log writing and concurrent administrator commission dashboard update.

Design of Software, System, Product, and/or Processes

Database Design

The Database Design of NutriMatch translates the Domain Class Diagrams into a concrete, implementation-ready relational structure on MySQL, encompassing the Database Schema (*Figure 34*) and the Data Dictionary (*Figure 35*).

The Database Schema comprises twenty-one interrelated tables organized across six functional groups: User and Authentication, Scheduling, Clinical NCP, Food Exchange, Communication and Resources, and Billing and Financial. The central users table anchors the structure through one-to-one extensions to `rnd_profiles` and `client_profiles`, with `rnd_client_relationships` serving as the foreign key hub for all clinical, financial, and communication records. The FNRI FCT API and USDA FoodData Central are consumed transiently, with only the plain-text `food_name` field persisted, maintaining RA 10173 data minimization compliance. The Data Dictionary provides a field-level specification for every column, documenting clinically significant constraints including bcrypt password hashing, WHO Asia-Pacific BMI classification thresholds, the composite unique constraint preventing duplicate care pairings, the immutability of `audit_log` rows, and the commission amount freezing mechanism in the invoices table.

System Architectural Design

The System Architectural Design is represented through five System Class Diagrams (*Figures 36 through 40*). *Figure 36* illustrates the RND-Client Matching cluster, with the `MatchingController` invoking the `RndProfileRepository` and the `RndClientRelationship` model managing pending, active, and discharged lifecycle states. *Figure 37* illustrates the NCP Documentation cluster, with the `NutritionalCalculationEngine` encapsulating the Mifflin-St Jeor BMR, TDEE, WHO Asia-Pacific BMI, and NRS-2002 algorithms. *Figure 38* illustrates the Meal Planning cluster, with the `MacronutrientAggregator` processing API responses in memory only and no nutrient payload written to the database. *Figure 39* illustrates the Billing cluster, with the `CommissionCalculator` freezing amounts at invoice creation and the `PaymentGatewayClient` triggering RA 10173-compliant audit logging on successful payment. *Figure 40* illustrates the Resources and Reminders cluster, with the

ReminderScheduler implemented as a Laravel scheduled command and the MessageController supporting RA 10173-compliant in-platform clinical messaging.

UI/UX Design

The UI/UX Design employs a teal and amber color palette with DM Serif Display for branding and DM Sans for body text across fully responsive, role-specific dashboards. A persistent vertical sidebar provides primary navigation, collapsing into a hamburger menu on mobile. The RND-Client matching interface (*Figure 41*) presents RND cards with credentials, ratings, consultation fee, and specialty badges, with a Request Consultation button opening a scheduling modal. The NCP module (*Figure 42*) is organized as a four-phase progressive disclosure interface, with Phase 1 displaying a live-updating panel showing BMI, BMR, TDEE, and NRS-2002, Phase 2 providing an auto-generated PES statement preview, and Phases 3 and 4 covering diet prescription and monitoring notes, with all phases supporting draft saving. The meal planning module (*Figures 43 and 44*) presents a condition-specific template selector with FNRI and USDA food search, six meal slots with real-time macro summary, and a printable client-facing plan. The resource repository (*Figures 45 and 46*) and automated reminder system enforce role-based access control, ensuring resources are visible only to their intended recipients. The billing interfaces (*Figures 47, 48, and 49*) provide role-appropriate views for Clients, Dietitians, and Administrators with color-coded payment status badges and commission tracking, with frozen commission amounts protecting against retroactive financial adjustments.

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Appendices

MONTH	Feb				Mar				Apr				May				Jun				Jul				Aug				Sep				Oct				Nov				Dec				
ACTIVITY	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4					
CAPSTONE TITLE BRAINSTORMING	Blue	Blue	Yellow																																										
CAPSTONE PROPOSAL		Blue	Blue	Yellow																																									
PROJECT CONTEXT			Blue	Yellow																																									
PURPOSE AND DESCRIPTION			Blue	Yellow																																									
OBJECTIVES				Blue	Blue	Yellow																																							
SCOPE AND LIMITATION				Blue	Blue	Yellow																																							
OBJECTIVES (REVISION)						Blue	Yellow																																						
REVIEW OF RELATIVE LITERATURE/STUDIES/SYSTEMS							Yellow																																						
REVIEW OF RELATIVE LITERATURE/STUDIES/SYSTEMS (REVISION)							Yellow																																						
METHODOLOGY								Blue	Blue	Blue	Yellow	Yellow																																	
REQUIREMENT GATHERING & REQUIREMENT ANALYSIS								Blue	Blue	Blue	Yellow	Yellow																																	
REQUIREMENT DOCUMENTATION								Blue	Blue	Blue	Yellow	Yellow																																	
SYSTEM DESIGN								Blue	Blue	Blue	Yellow	Yellow																																	
METHODOLOGY (REVISION)									Blue	Blue	Yellow	Yellow																																	
REQUIREMENT GATHERING & REQUIREMENT ANALYSIS (REVISION)									Blue	Blue	Yellow	Yellow																																	
REQUIREMENT DOCUMENTATION (REVISION)									Blue	Blue	Yellow	Yellow																																	
SYSTEM DESIGN (REVISION)												Yellow	Yellow	Yellow																															
SYSTEM DEVELOPMENT																																													
TESTING & EVALUATION																																													
DOCUMENTATION																																													
FINAL DEFENSE																																													
REVISION & SUBMISSION																																													

Above are the legends of activity indication where **BLUE** indicates as starting or not finished and **YELLOW** as on progress or finished.

Figure 3. Calendar of Activities

Specification	Device 1	Device 2
Processor (CPU)	Intel® Core™ i5-6400 @ 2.70 GHz	AMD Ryzen 5 PRO 3500U @ 2.10 GHz
Operating System	Windows 10 Pro	Windows 11 Home
Memory (RAM)	16 GB RAM 3200 MHz	16 GB RAM
Storage	1TB HDD / 512GB SSD	1TB HDD / 512GB SSD
Monitor / Display	27-inch Monitor Full HD 1920x1080, 144Hz	16-inch HD 1920x1080, 144Hz

Figure 4. Hardware Specification

<i>Software</i>	<i>Stable Release</i>
<i>Laravel</i>	<i>11.x</i>
<i>Nuxt.js</i>	<i>3.10+</i>
<i>MySQL</i>	<i>8.0+</i>
<i>Visual Studio Code</i>	<i>1.85+</i>
<i>Postman</i>	<i>11.x</i>
<i>GitHub</i>	<i>Web-based Platform (Latest)</i>
<i>Figma</i>	<i>Web-based Platform (Latest)</i>

Figure 5. Software Stable Release

Question / Item	Respondent 1 (Jimenez)	Respondent 2 (Forones)
Section 1: Respondent Profile		
Name (optional):	Joerine Trexie Jimenez	Gyle Francis Forones
Years of experience as an RND:	0–2 years	0–2 years
Practice setting (check all that apply):	Hospital	Hospital
Do you currently conduct online consultations?	Yes	No
Section 2: Current Workflow		
How do you currently manage patient records? (Check all that apply)	Excel/Spreadsheets, Combination	Paper-based, Excel/Spreadsheets, Combination
Describe your step-by-step process during a typical consultation:	Conduct interviews for medical history, age, weight, height, dietary habits, allergies, religion, family history, and laboratory results; compute diet prescription and create tailored meal plan; monitor patients and check actual meals against the plan; answer follow-up questions on acceptable food items.	Assess the patient's nutritional status using paper-based assessment forms; conduct consultation proper and counsel the patient; provide interventions and continuously monitor nutritional status.
How do you compute BMI, BMR, TDEE, and Nutritional Risk? (Check all that apply)	Manually, Using calculators/tools	Manually, Using calculators/tools
How do you create meal plans? (Check all that apply)	Manual computation, FNRI Food Exchange Lists (printed)	Manual computation, FNRI Food Exchange Lists (printed), Software/app
How do you track patient progress? (Check all that apply)	Notes, Messaging apps	Notes, Excel, Messaging apps
How do you communicate with patients after consultation? (Check all that apply)	Messenger	Messenger, Email
Section 3: Challenges and Pain Points		
What are the biggest challenges in your current workflow?	Progress monitoring patients lose consistency halfway through the meal plan.	Storing data conveniently and monitoring patient progress.
Which tasks take the most time? (Check all that apply)	Meal planning, Patient monitoring	Meal planning, Patient monitoring
Do you experience any of the following issues? (Check all that apply)	Difficulty tracking progress, Miscommunication with patients	Lost patient records, Repetitive manual computations, Difficulty tracking progress, Miscommunication with patients, Data privacy concerns
Section 4: Proposed Feature Importance (1 = Not Important → 5 = Extremely Important)		
Automated BMI, BMR, TDEE computation	Very Important (5)	Very Important (5)
Digital patient records	Very Important (5)	Very Important (5)
Nutrition Care Process (NCP) documentation	Very Important (5)	Very Important (5)
PES statement generator	Important (4)	Very Important (5)
Disease-specific meal planning	Very Important (5)	Very Important (5)
FNRI-based food database	Very Important (5)	Very Important (5)
Client progress dashboard	Very Important (5)	Very Important (5)

Automated report generation	Very Important (5)	Very Important (5)
Messaging system within platform (including video conferencing)	Very Important (5)	Very Important (5)
Online payment system	Very Important (5)	Very Important (5)
Reminder/notification system	Very Important (5)	Very Important (5)
Section 5: Client-RND Matching Importance (1 = Not Important → 5 = Extremely Important)		
Specialty (e.g., diabetes, renal)	5	5
Language preference	4	4
Consultation type (online/face-to-face)	5	5
Section 6: Consultation Features Needed		
What consultation features do you need? (Check all that apply)	Video call, Chat, File sharing (lab results, diet plans)	Video call, Chat, File sharing (lab results, diet plans)
Section 7: Open-Ended Responses		
What features do you wish existed in a nutrition system?	Progress monitoring for patients; a food intake calculator where patients encode their daily food items and get equivalent nutrient content.	An app or website that is easily accessible for handling and processing nutrition care.
If a system like NutriMatch is developed, would you use it?	Yes	Yes
Additional suggestions:	(none)	None so far.

Figure 6. Questionnaire

A. Functional Model - Use Case Diagram

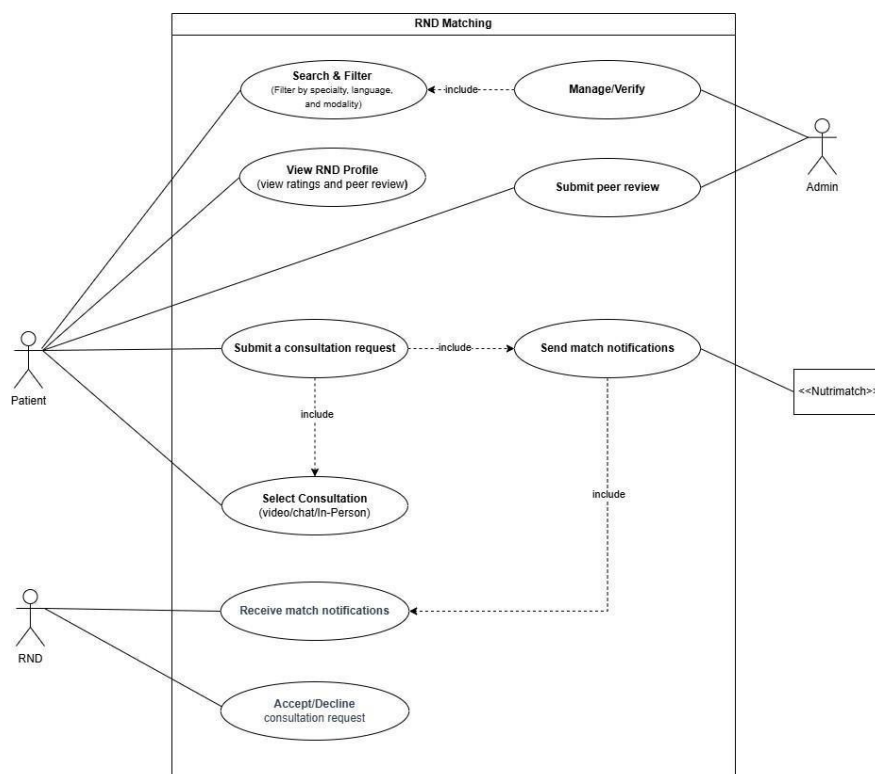


Figure 7. RND-Client Matching

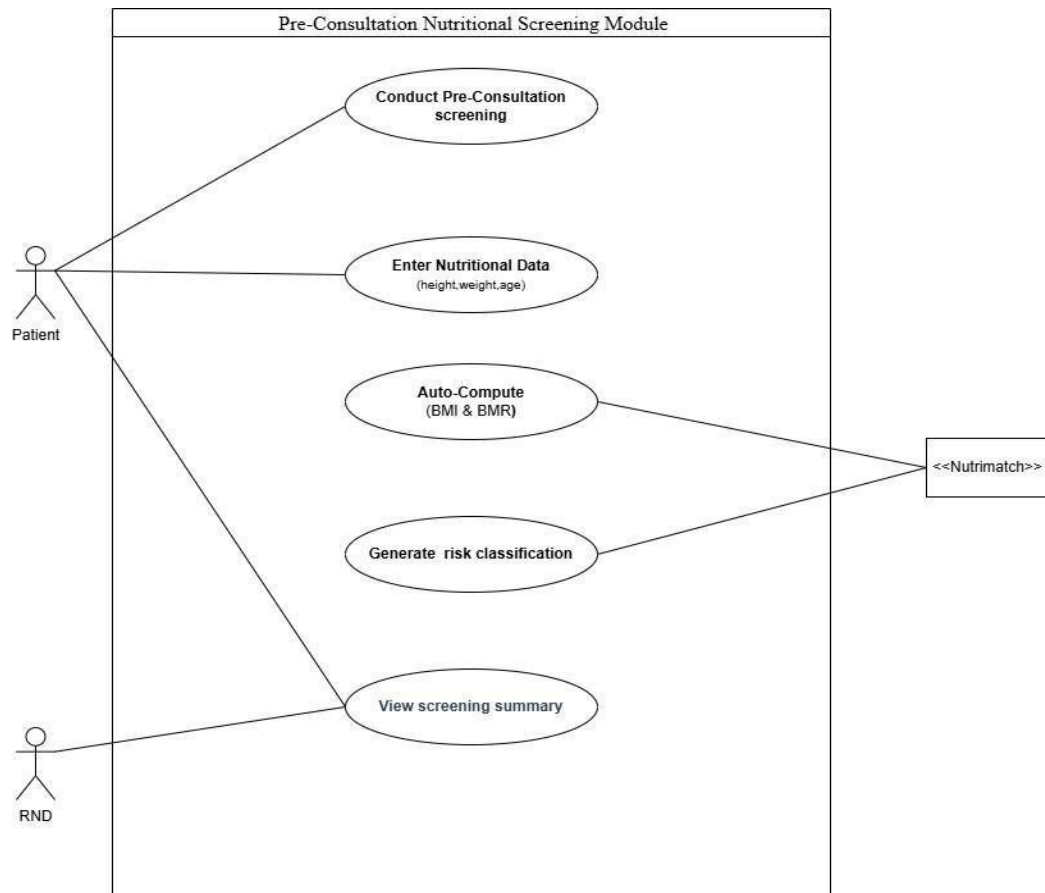


Figure 8. Pre-Consultation Module

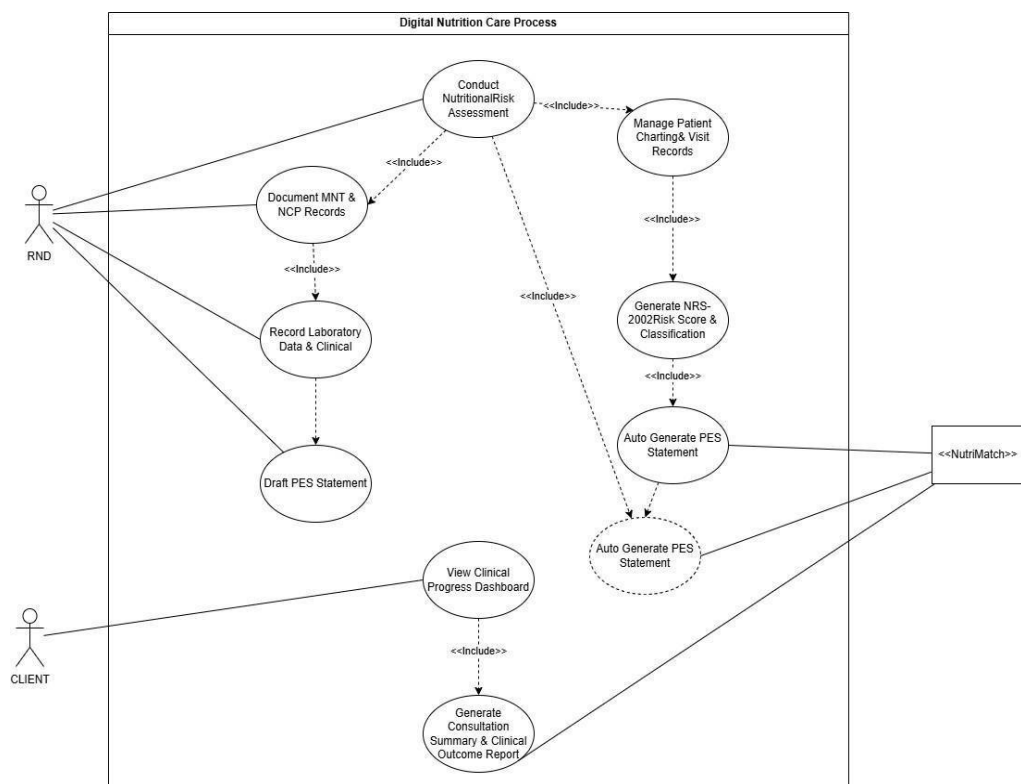


Figure 9. Digital Nutrition Care Process Module

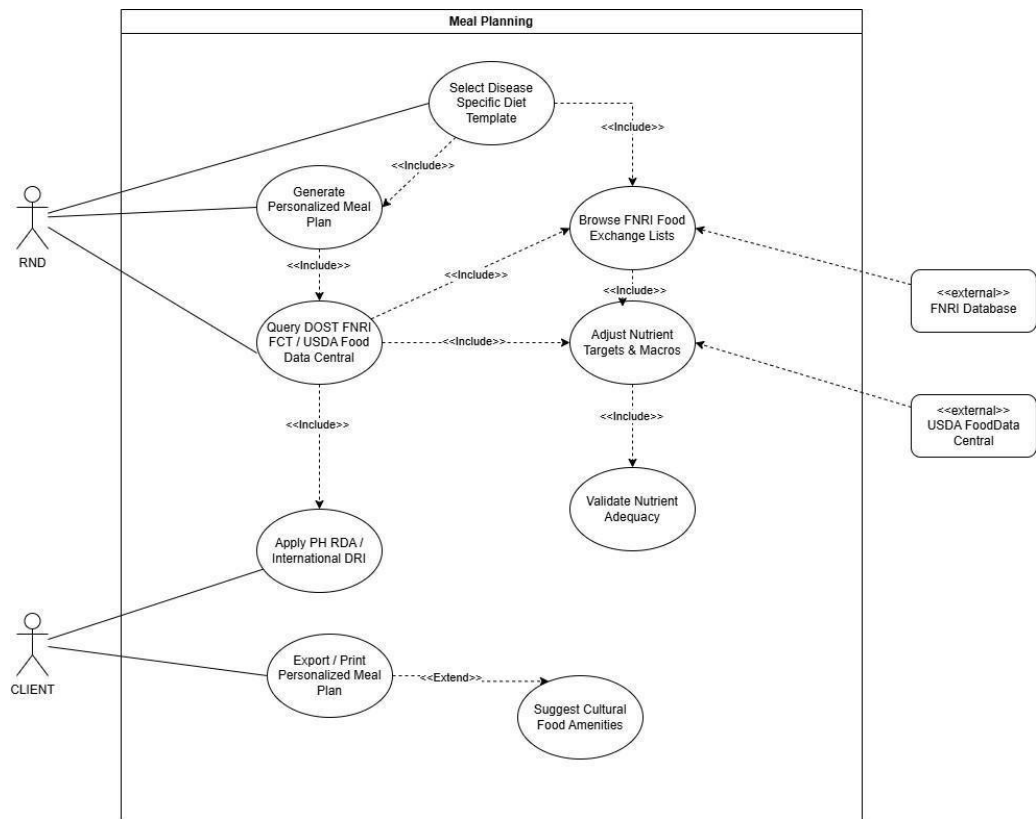


Figure 10. Meal Planning Module

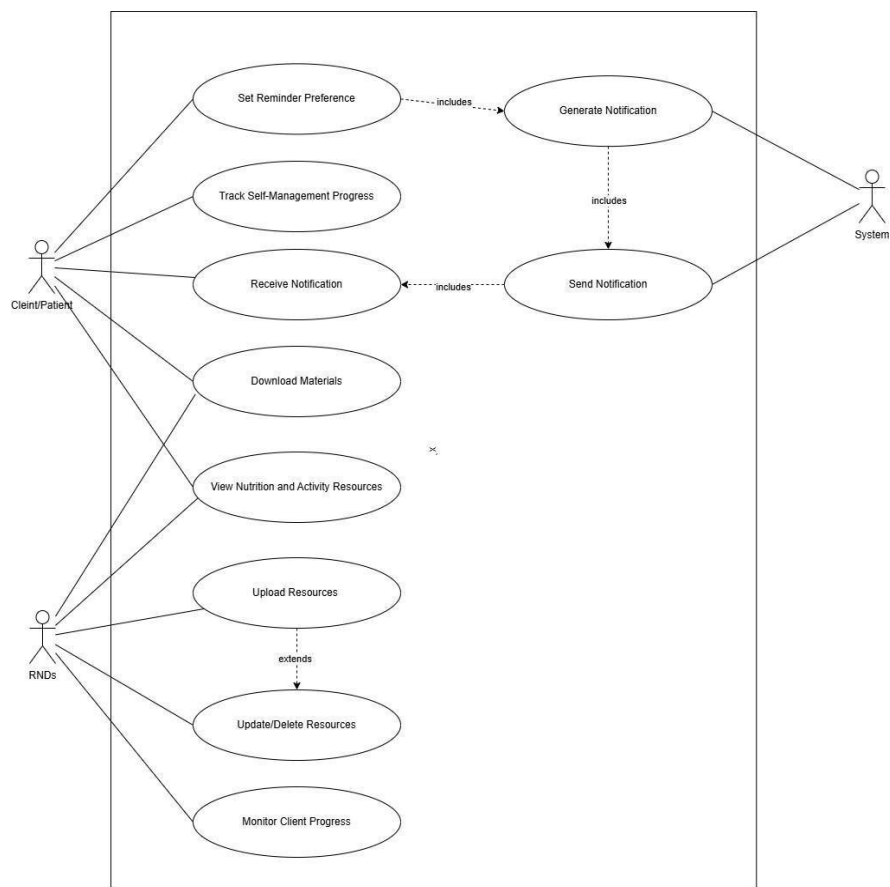


Figure 11. Digital Repository and Automated Reminder Module

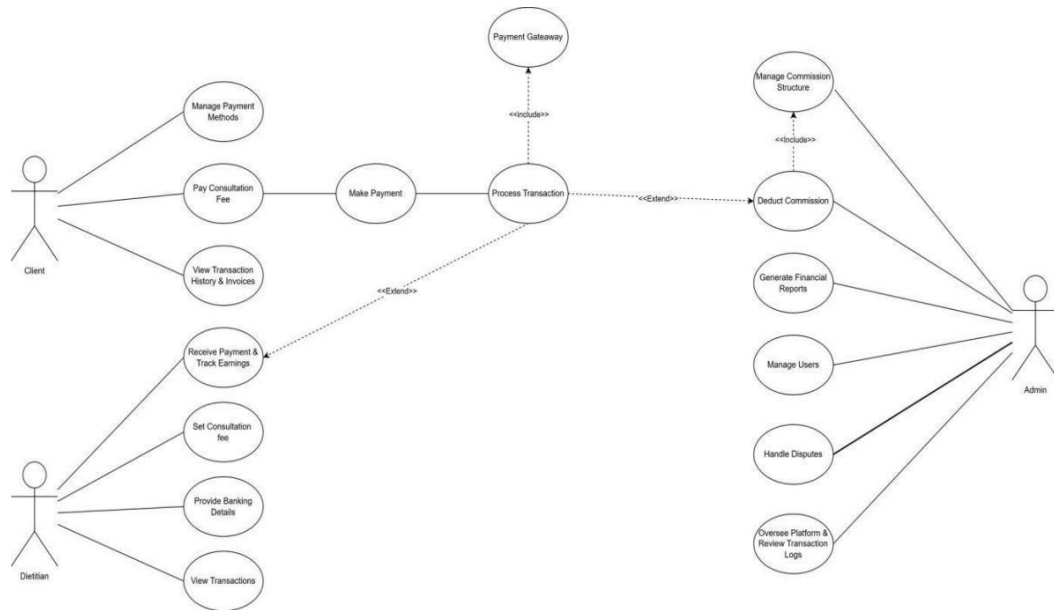


Figure 12. Billing Module

B. Process Model - BPMN Diagram

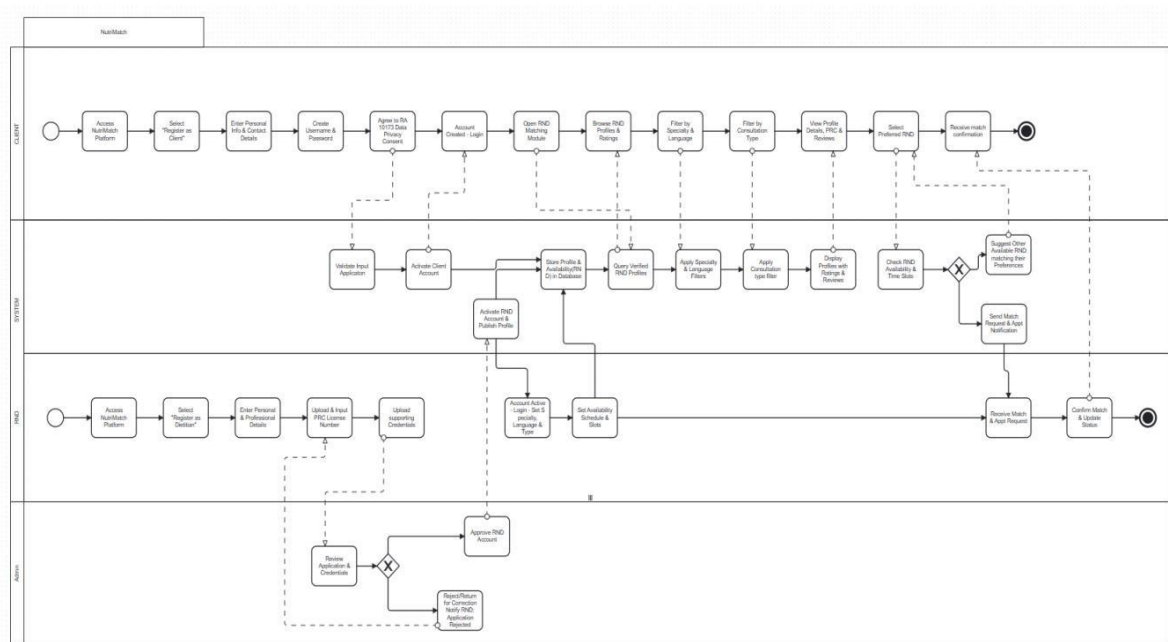


Figure 13. RND-Client Matching

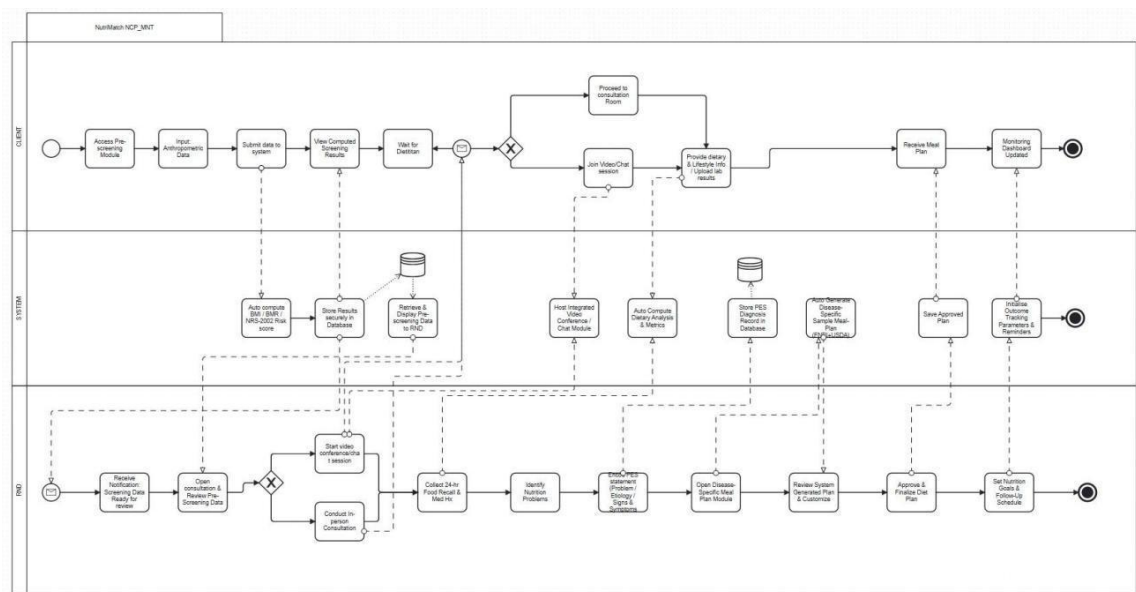


Figure 14. Medical Nutrition Therapy - Nutrition Care Process

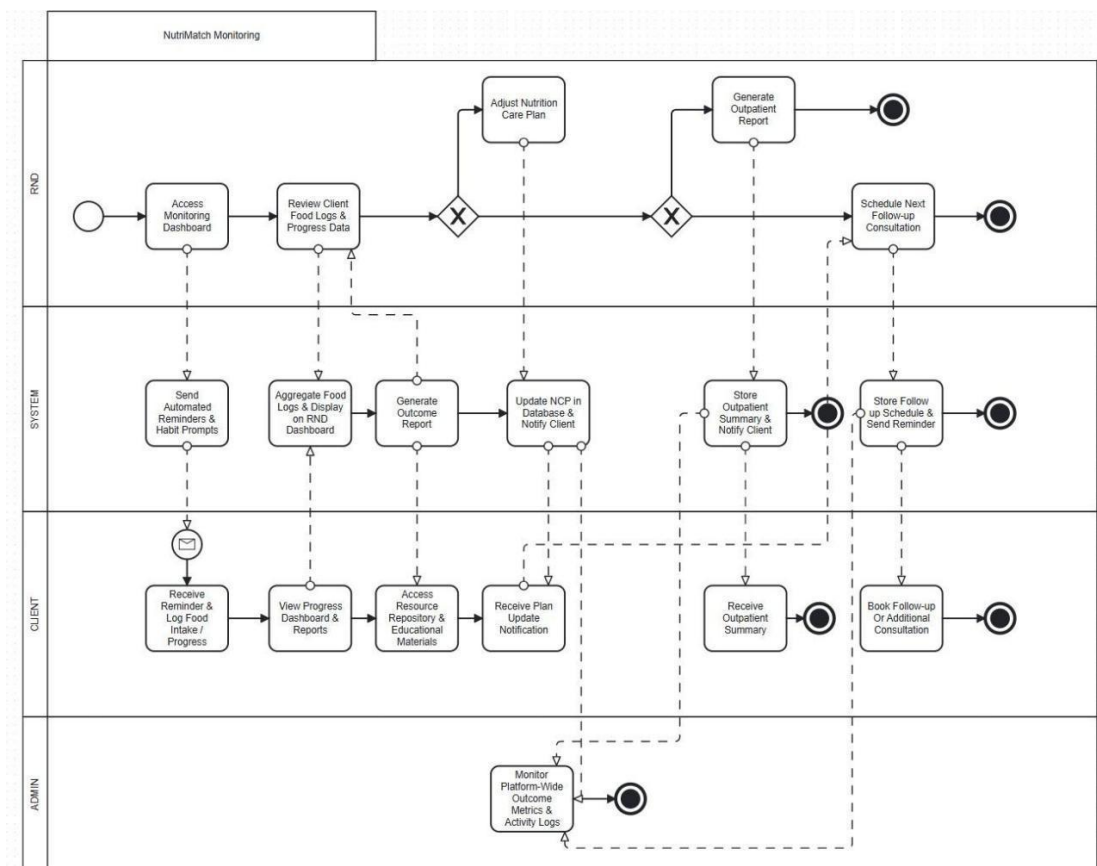


Figure 15. Monitoring and Follow-up BPMN Diagram

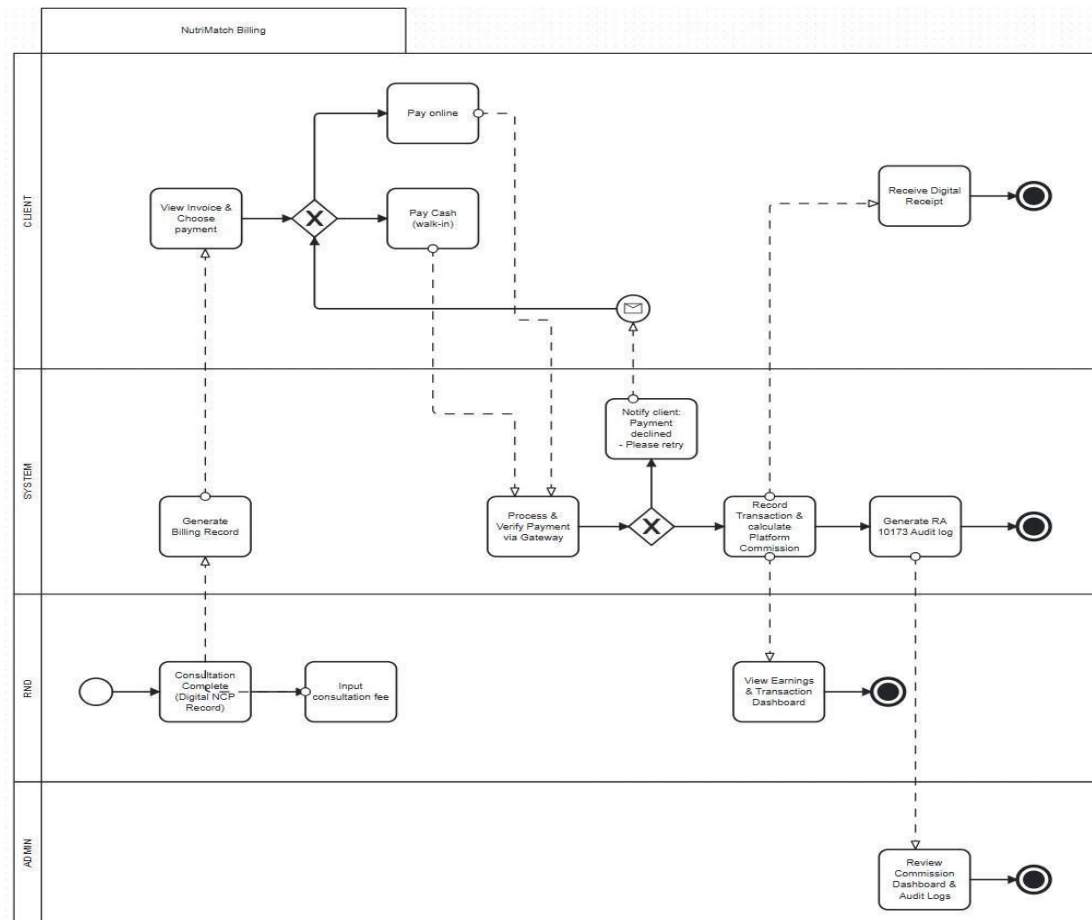


Figure 16. Billing BPMN Diagram

Activity Diagram

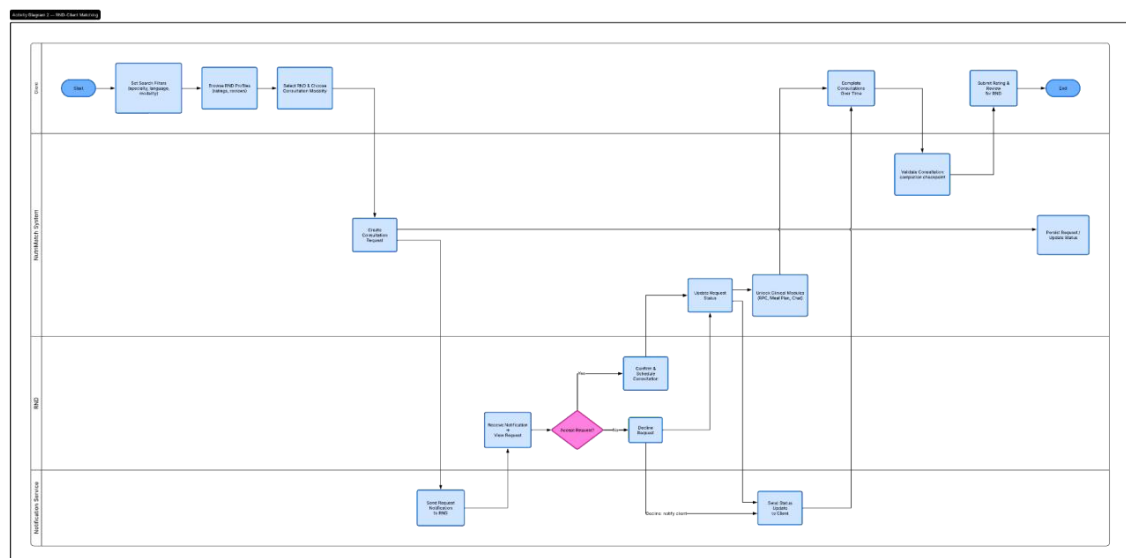


Figure 17. RND-Client Matching Activity Diagram

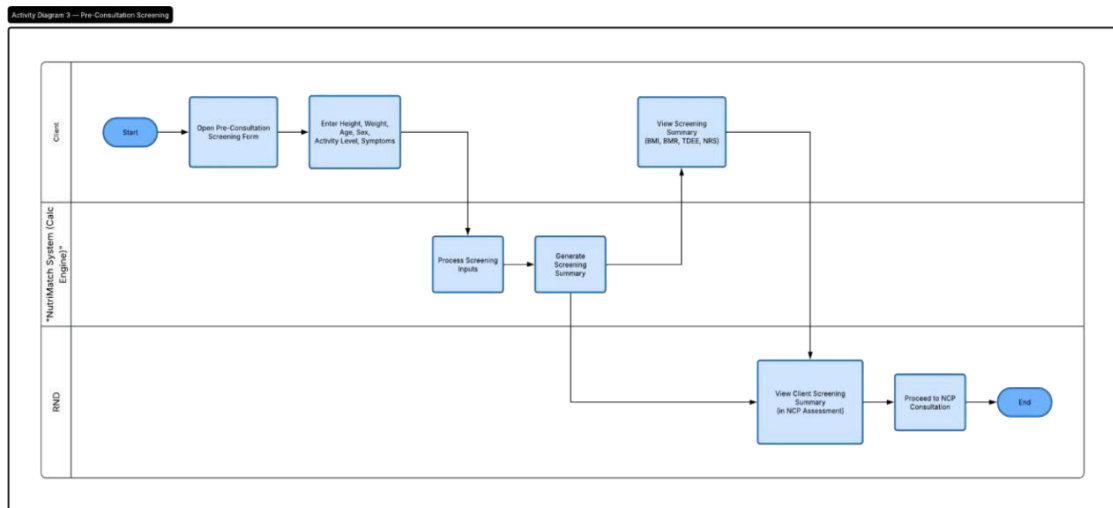


Figure 18. Pre-Consultation Screening Activity Diagram

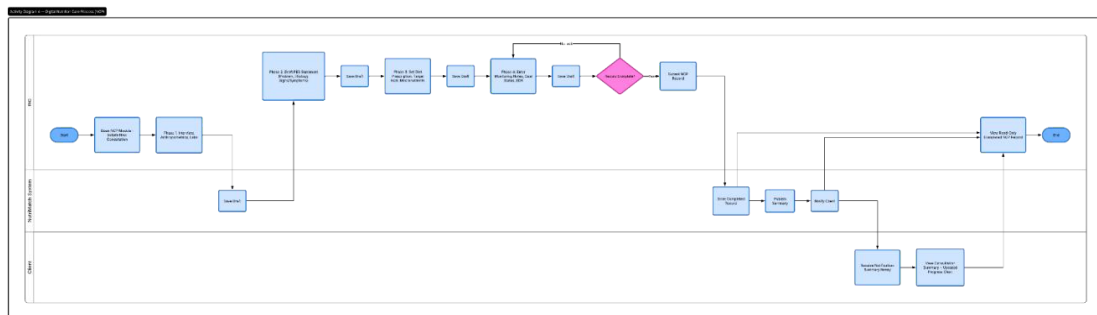


Figure 19. Digital Nutrition Care Process Activity Diagram

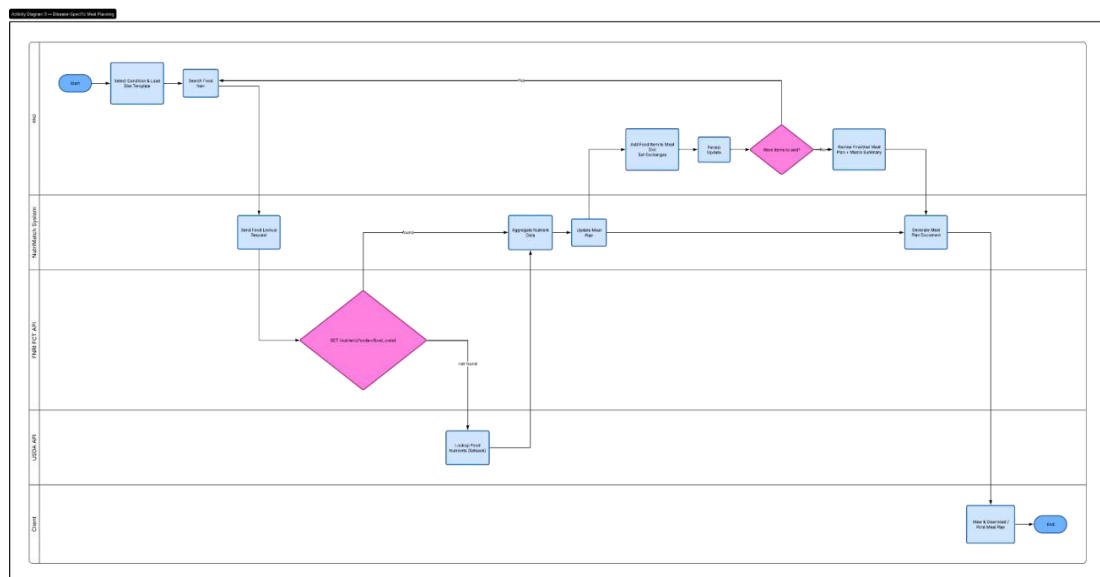


Figure 20. Disease Specific Meal-Planning Activity Diagram

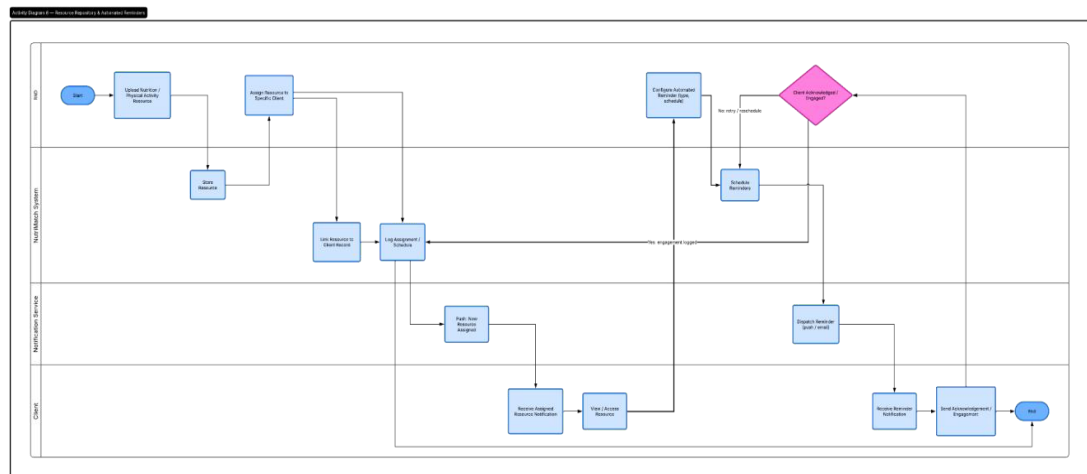


Figure 21. Resource Repository and Automated Reminder Activity Diagram

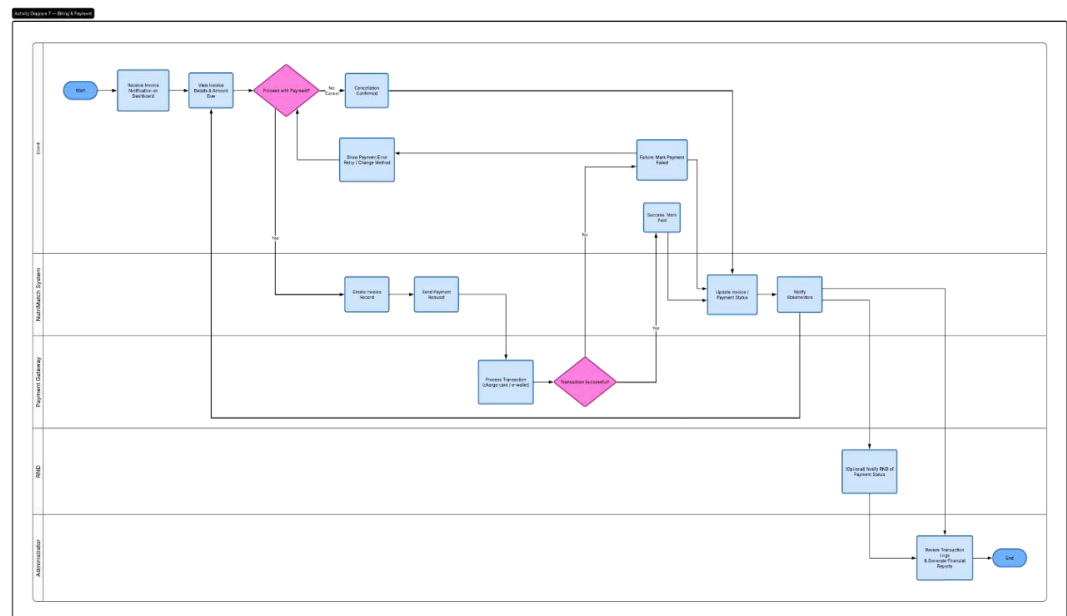


Figure 22. Billing and Payment Activity Diagram

B. Data Model - Domain Class Diagram

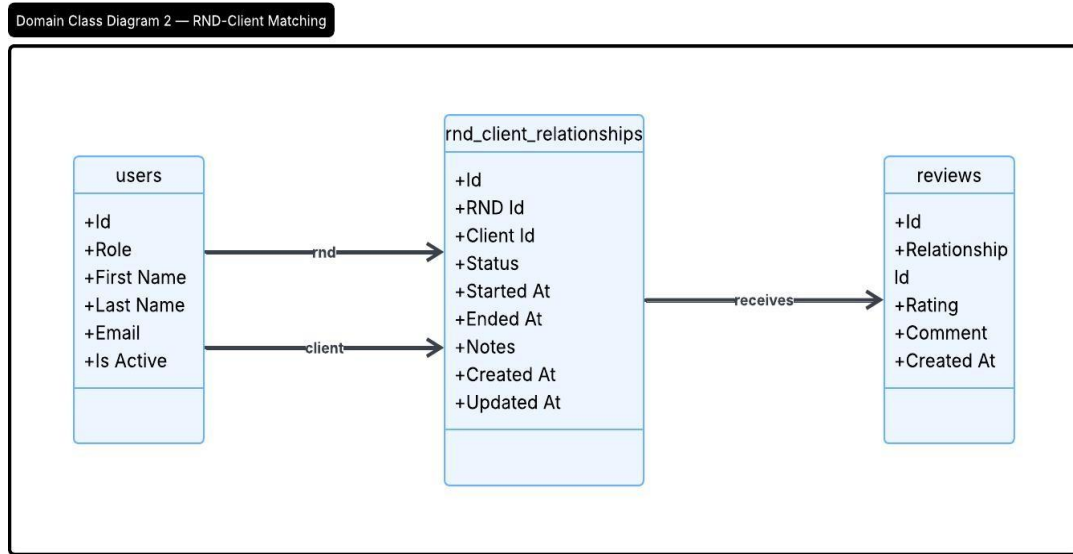


Figure 23. RND Client Matching

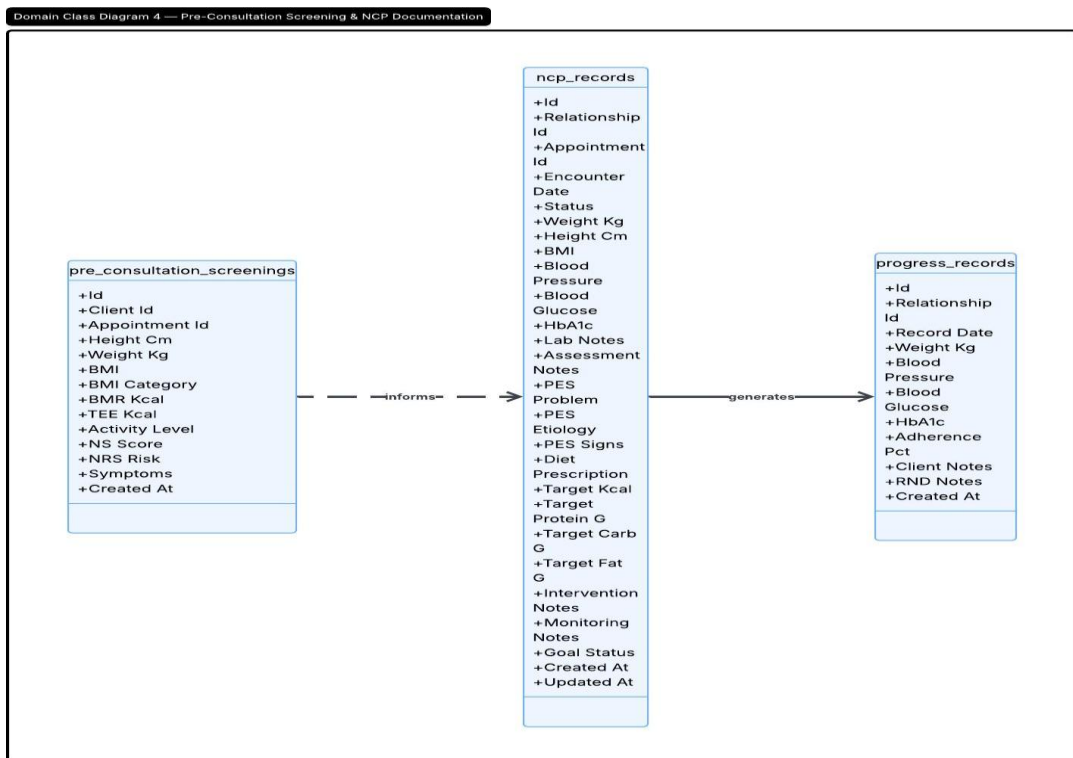


Figure 24. Pre-Consultation & NCP Documentation

Domain Class Diagram 5 — Meal Planning

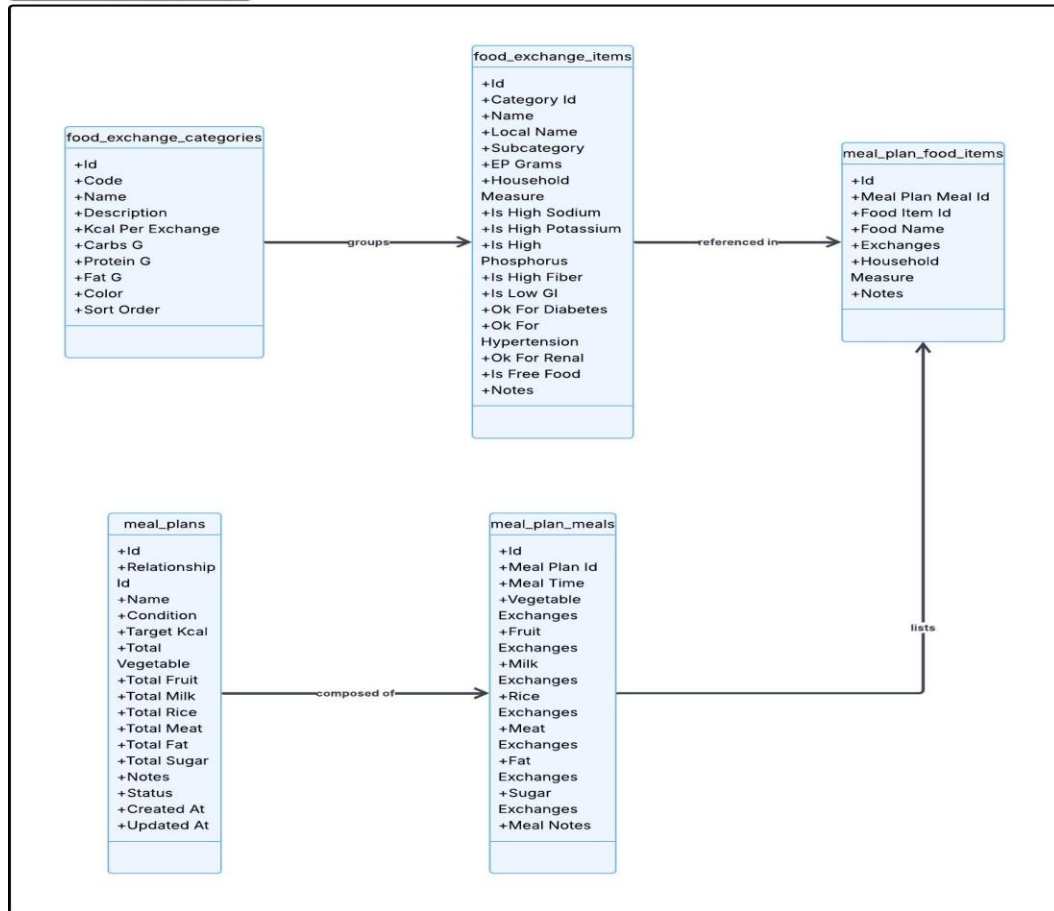


Figure 25. Meal Planning

Domain Class Diagram 6 — Billing & Invoicing

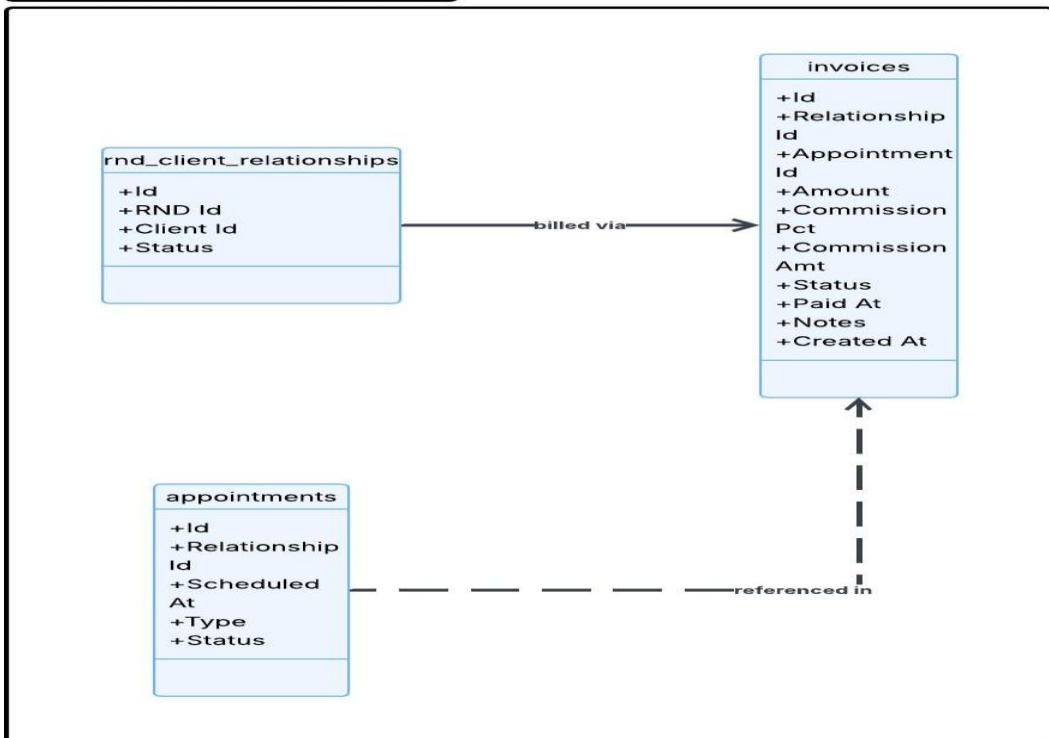


Figure 26. Billing & Invoicing

Domain Class Diagram 7 — Communication, Resources & Reminders

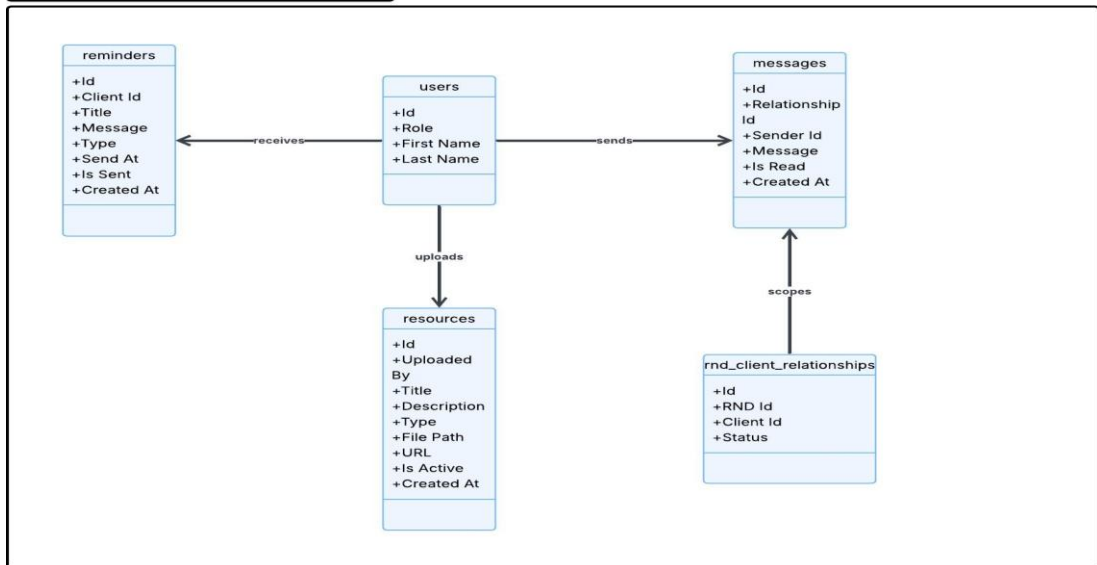


Figure 27. Communication, Resources & Reminders

C. Behavioral Model - Sequence Diagram

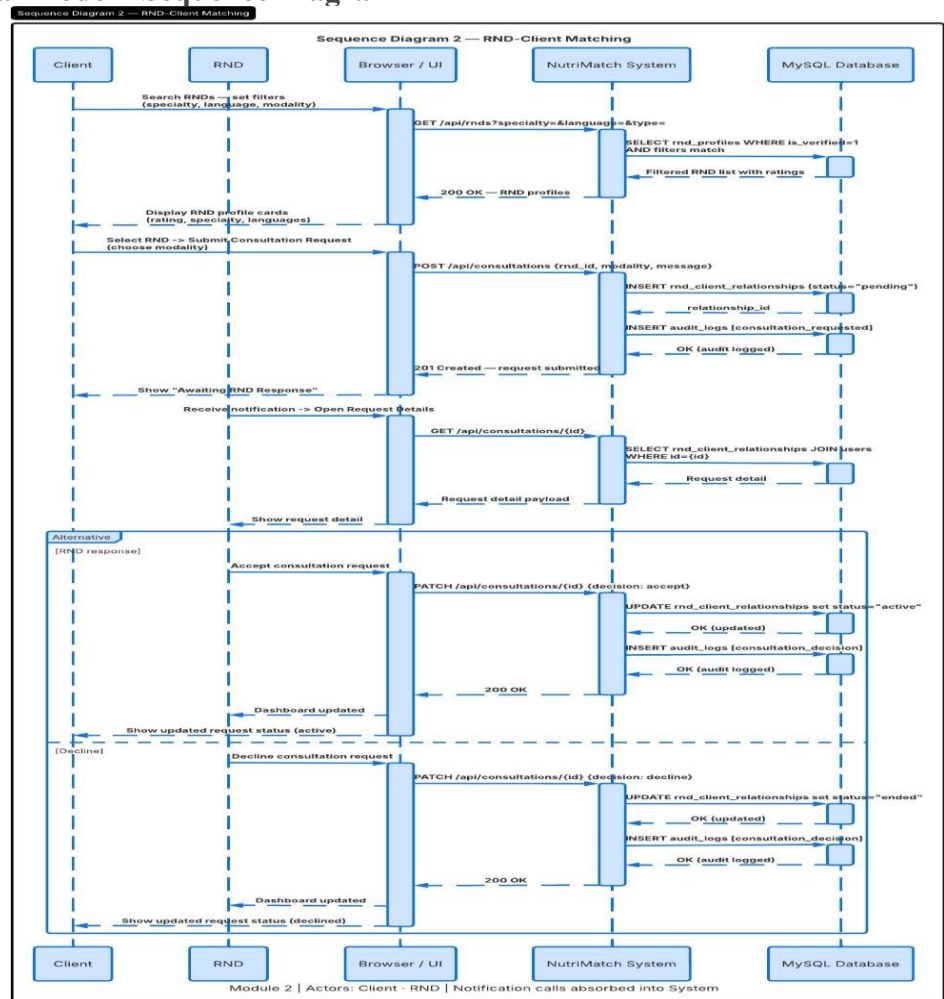


Figure 28. RND Client Matching

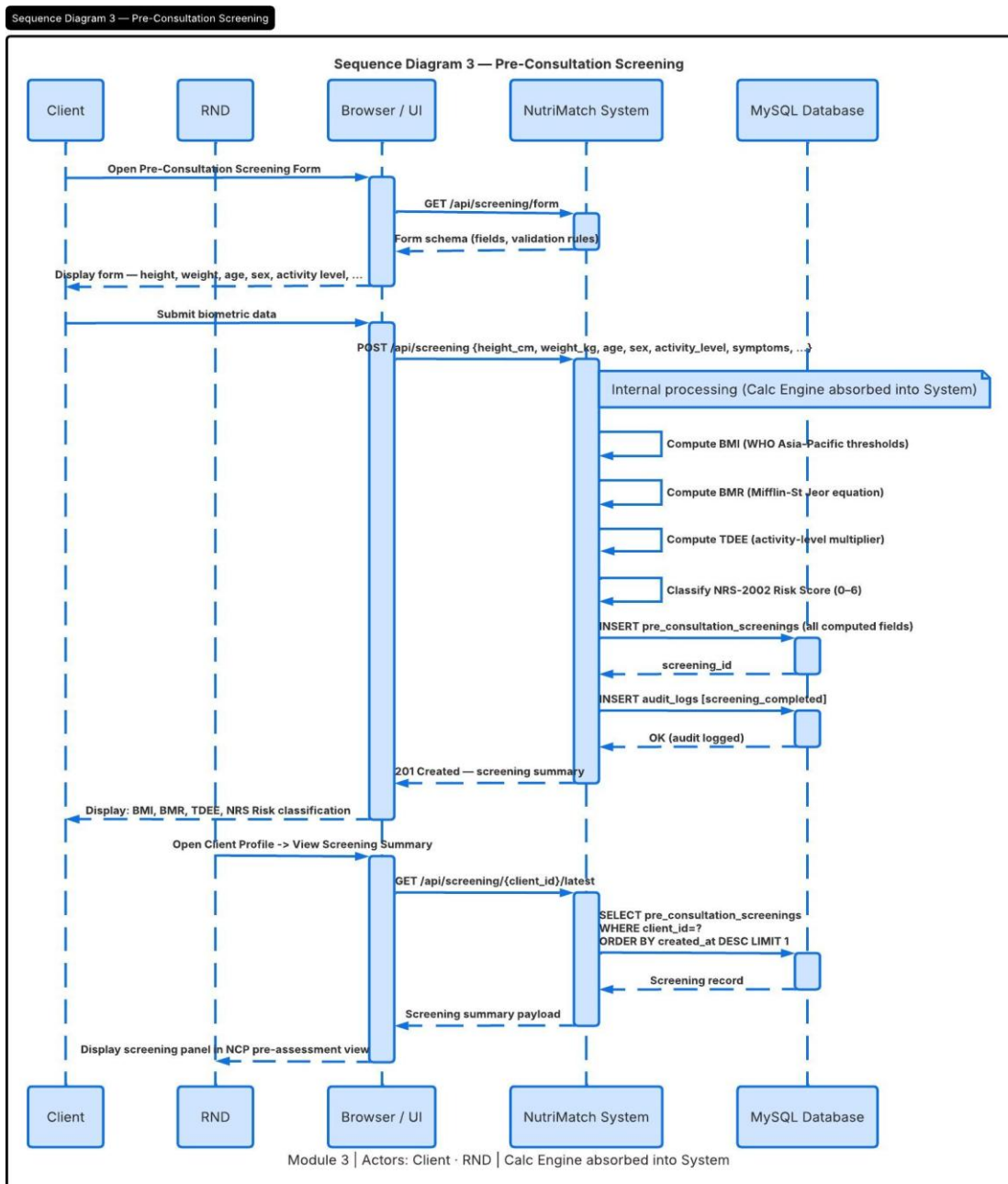


Figure 29. Pre-Consultation Screening

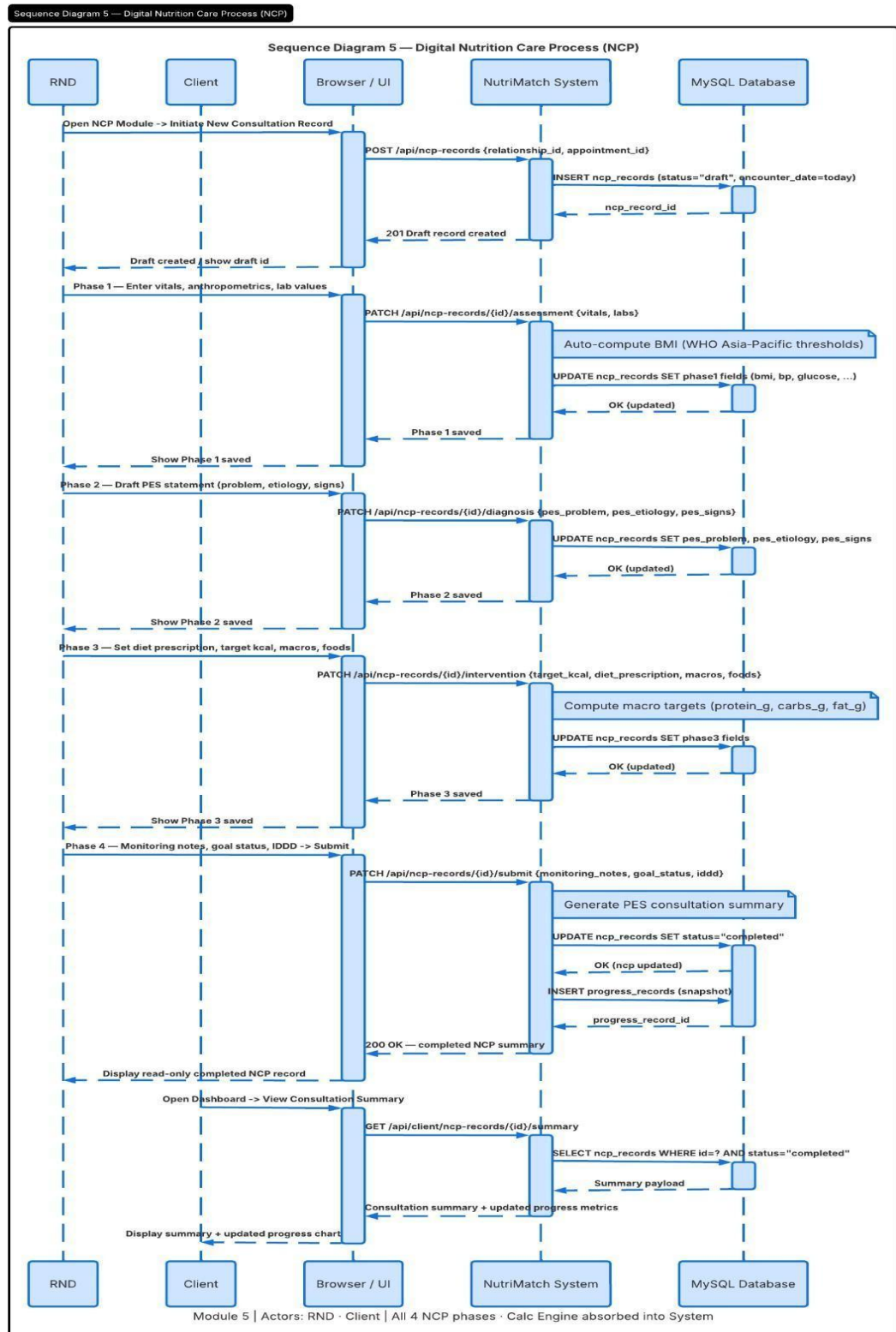


Figure 30. Digital Nutrition Care Process

Sequence Diagram 6 — Disease-Specific Meal Planning

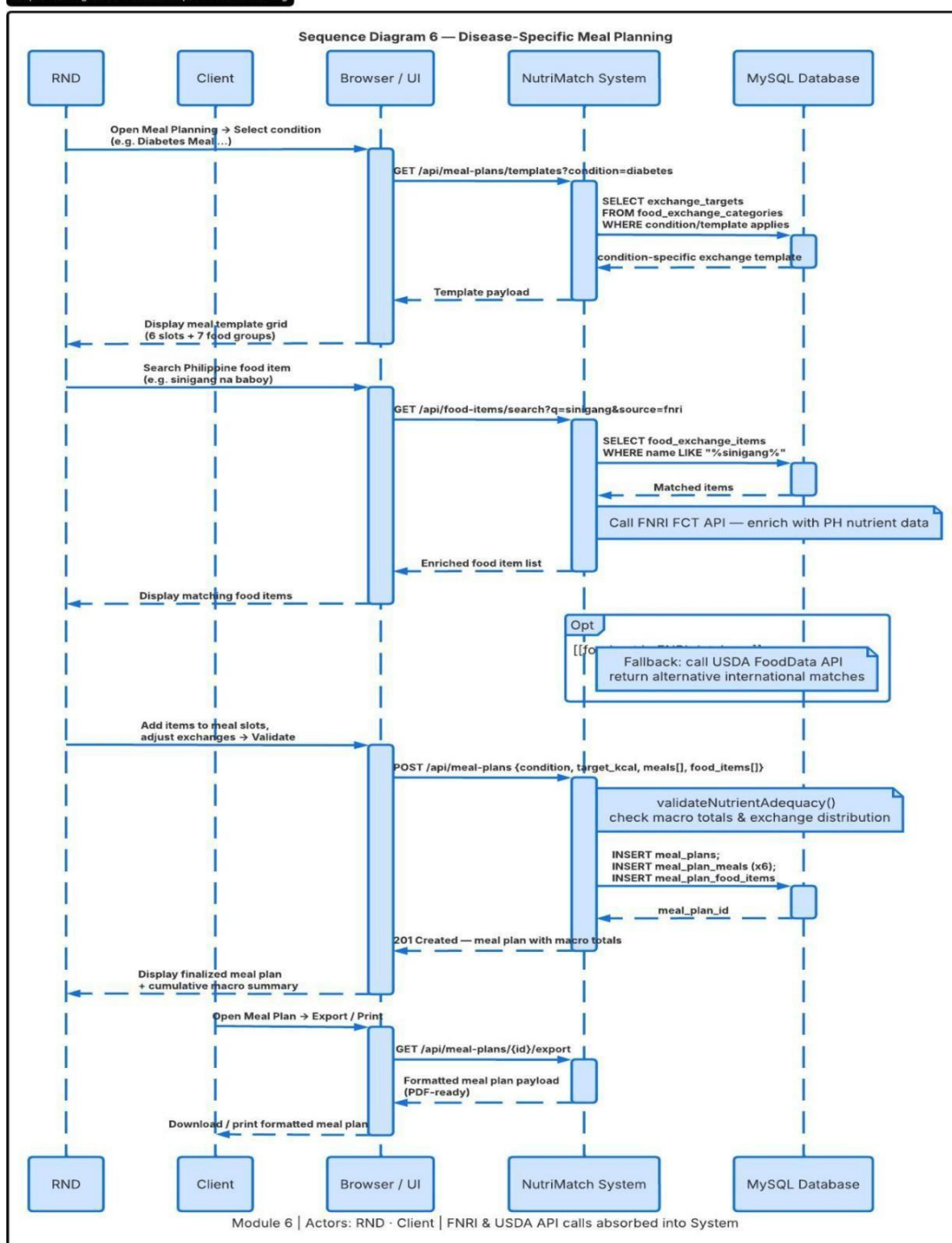


Figure 31. Disease-Specific Meal Planning

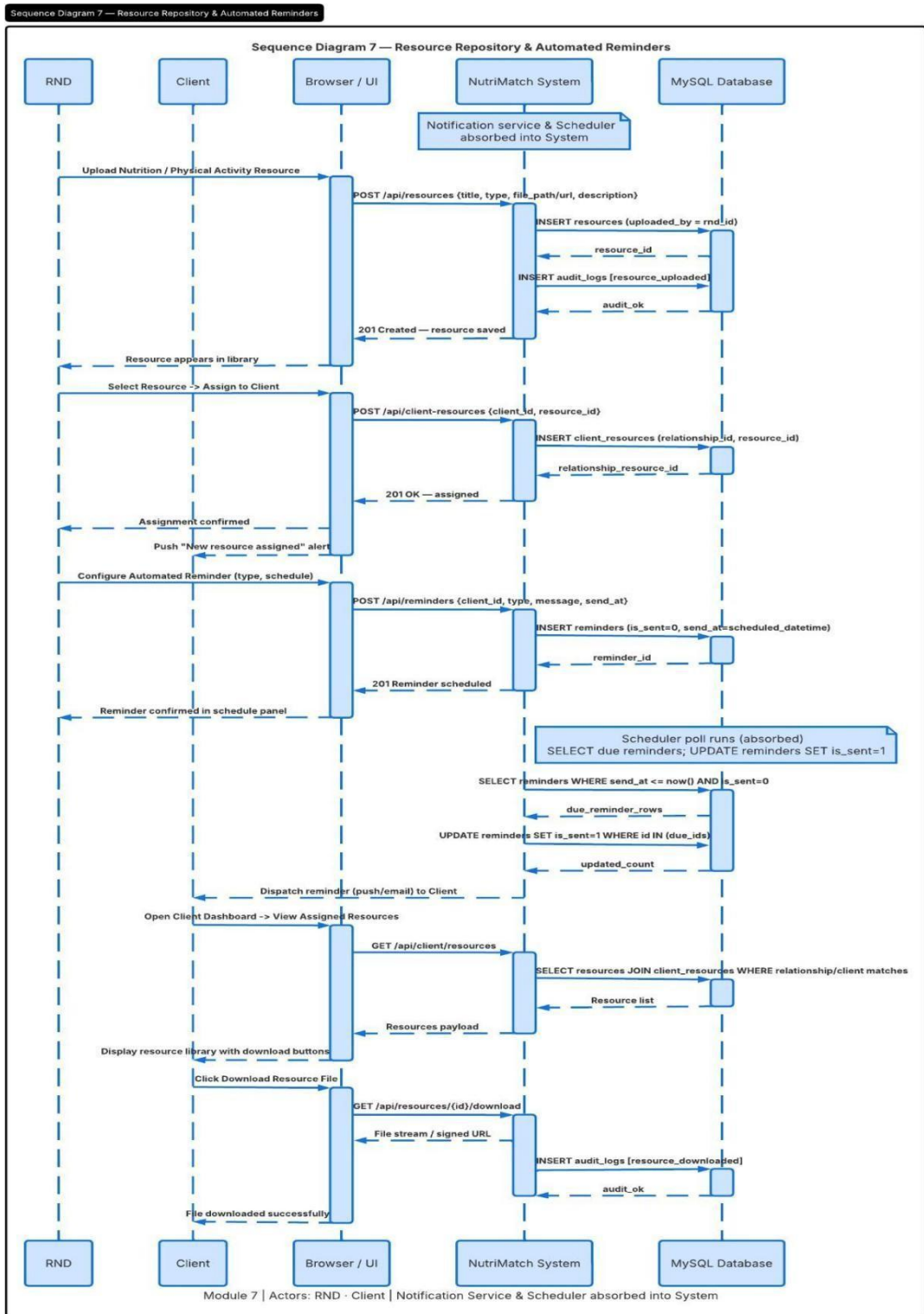


Figure 32. Resource Repository & Automated Reminders

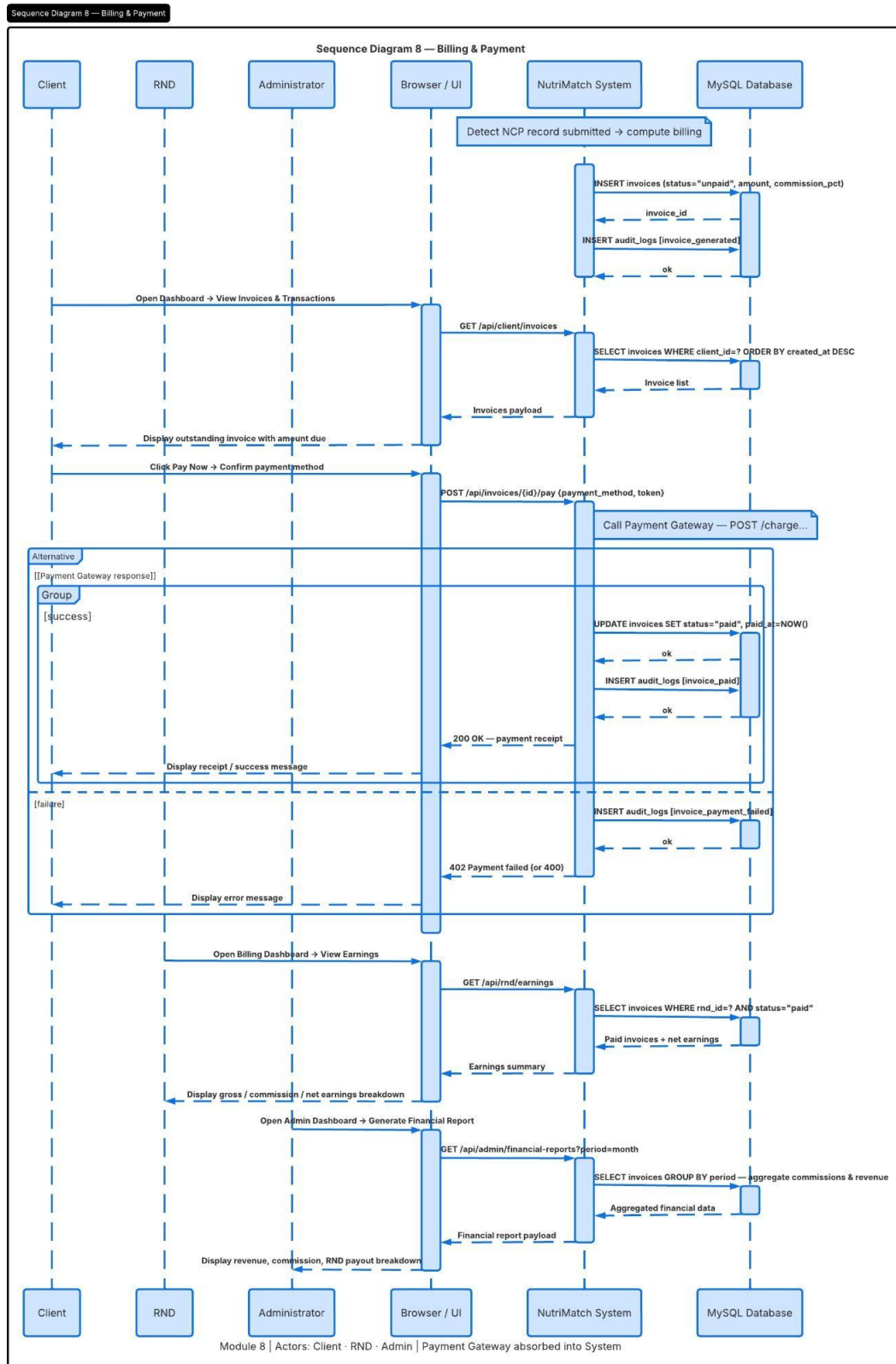


Figure 33. Billing & Payment

Database Schema

nutrimatch — Entity Relationship Diagram

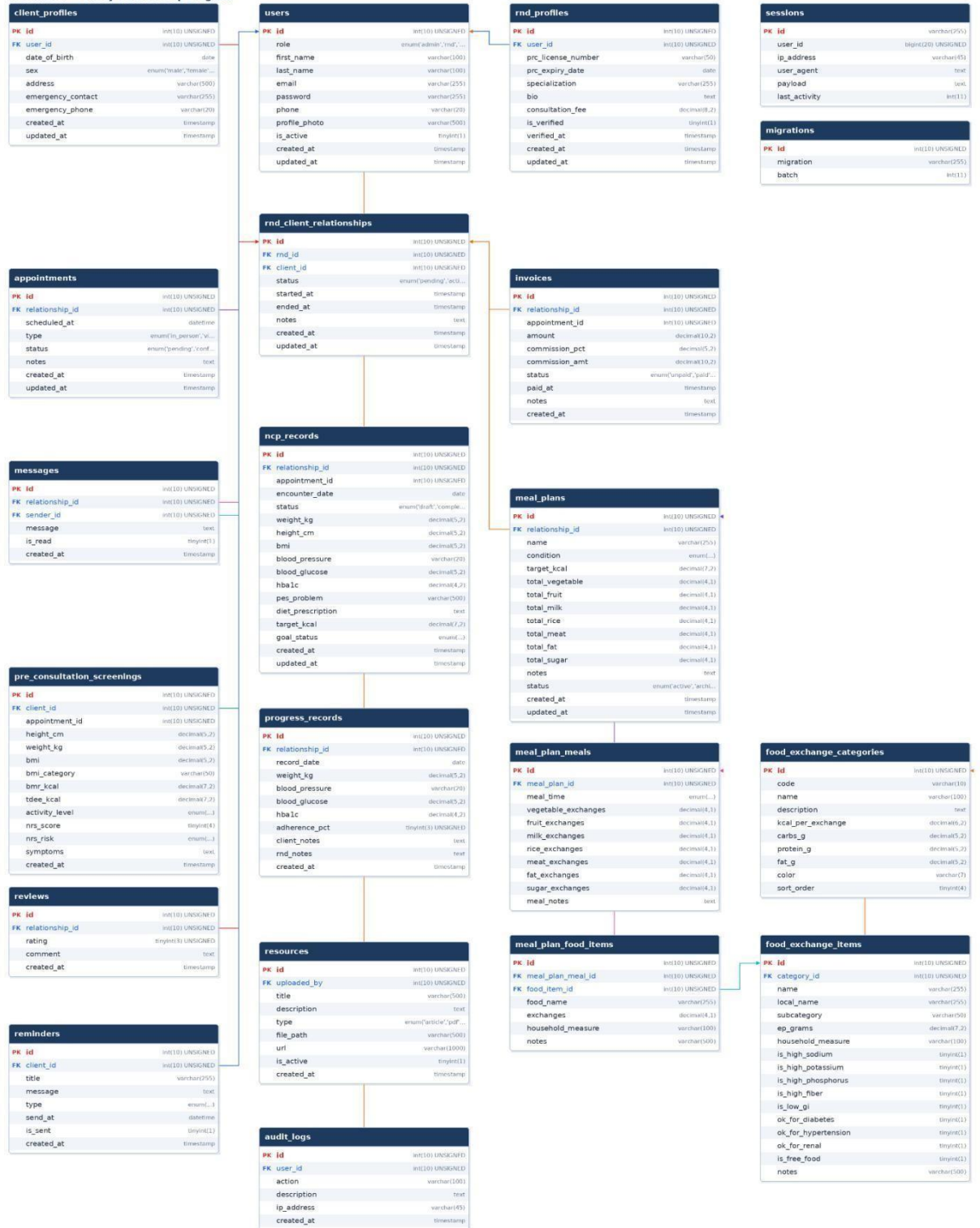


Figure 34. Database Schema

Data Dictionary

Key Notation	
PK — Primary Key — unique row identifier	FK — Foreign Key — references another table
UK — Unique Key — enforces uniqueness	NOT NULL — Field is mandatory; must have a value
NULL — Field is optional; may be left empty	DEFAULT — System-assigned default value

USER & AUTHENTICATION			
01. users			
Central identity table shared by all three roles (admin, rnd, client). Stores credentials, contact info, and role discriminator. Role-specific data lives in rnd_profiles or client_profiles.			
Column Name	Data Type	Constraints	Description
id	INT UNSIGNED	PK · AUTO_INCREMENT · NOT NULL	Surrogate primary key. Auto-assigned integer unique across all users.
role	ENUM	NOT NULL	Role discriminator: 'admin', 'rnd', or 'client'. Determines access scope.
first_name	VARCHAR(100)	NOT NULL	User's first name.
last_name	VARCHAR(100)	NOT NULL	User's last name.
email	VARCHAR(255)	NOT NULL · UK	Login email address. Must be unique across all users.
password	VARCHAR(255)	NOT NULL	bcrypt-hashed password. Plain-text passwords are never stored.
phone	VARCHAR(20)	NULL	Optional contact number for SMS reminders and emergency lookup.
profile_photo	VARCHAR(500)	NULL	File path or URL to the user profile photo. NULL if not uploaded.
is_active	TINYINT(1)	NOT NULL · DEFAULT 1	Soft-disable flag. 1 = active; 0 = deactivated. Deactivated users cannot log in but records are retained for audit.
created_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Record creation timestamp.
updated_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Timestamp of last record update.

USER & AUTHENTICATION 02. rnd_profiles			
Extends users for the RND role. Stores professional credentials, PRC license details, specialization, and consultation fee. One-to-one with users.			
Column Name	Data Type	Constraints	Description
id	INT UNSIGNED	PK · AUTO_INCREMENT · NOT NULL	Surrogate primary key.
user_id	INT UNSIGNED	NOT NULL · UK · FK → users.id	Foreign key to the parent user. UNIQUE enforces one-to-one relationship.
prc_license_number	VARCHAR(50)	NOT NULL · UK	PRC license number. Must be unique; cross-referenced by admin against the PRC verification portal.
prc_expiry_date	DATE	NULL	PRC license expiry date. Used to flag upcoming renewals.
specialization	VARCHAR(255)	NULL	Clinical specialization area (e.g., Renal Nutrition, Pediatric Nutrition).
bio	TEXT	NULL	Short professional biography displayed on the RND public profile.
consultation_fee	DECIMAL(8,2)	NOT NULL · DEFAULT 0.00	Consultation fee set by the RND. Applied automatically during invoice generation.
is_verified	TINYINT(1)	NOT NULL · DEFAULT 0	Admin verification flag. 0 = pending; 1 = verified. Only verified RNDs appear in client-facing search results.
verified_at	TIMESTAMP	NULL	Timestamp when the admin approved the RND application.
created_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Record creation timestamp.
updated_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Timestamp of last record update.

USER & AUTHENTICATION 03. client_profiles			
Extends users for the client role. Holds demographic and emergency contact data required for clinical assessment (BMI, age-based BMR). One-to-one with users.			
Column Name	Data Type	Constraints	Description

id	INT UNSIGNED	PK · AUTO_INCREMENT · NOT NULL	Surrogate primary key.
user_id	INT UNSIGNED	<i>NOT NULL · UK · FK → users.id</i>	Foreign key to the parent user. UNIQUE enforces one-to-one relationship.
date_of_birth	DATE	NULL	Client date of birth. Used to calculate age for BMR computation.
sex	ENUM	NULL	Biological sex: 'male' or 'female'. Required for Mifflin-St Jeor BMR formula.
address	VARCHAR(500)	NULL	Home or mailing address of the client.
emergency_contact	VARCHAR(255)	NULL	Full name of the emergency contact person.
emergency_phone	VARCHAR(20)	NULL	Contact number of the emergency contact person.
created_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Record creation timestamp.
updated_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Timestamp of last record update.

SCHEDULING 04. rnd_client_relationships			
Junction table formalizing the care relationship between one RND and one client. Central hub — appointments, NCP records, meal plans, progress records, messages, invoices, and reviews all reference this table.			
Column Name	Data Type	Constraints	Description
id	INT UNSIGNED	PK · AUTO_INCREMENT · NOT NULL	Surrogate primary key.
rnd_id	INT UNSIGNED	<i>NOT NULL · FK → users.id</i>	Foreign key to the RND user. Forms a unique pair with client_id.
client_id	INT UNSIGNED	<i>NOT NULL · FK → users.id</i>	Foreign key to the client user. A client may have at most one relationship per RND.
status	ENUM	NOT NULL · DEFAULT pending	Lifecycle status: 'pending', 'active', or 'discharged'.
started_at	TIMESTAMP	NULL	Timestamp when the relationship became active.
ended_at	TIMESTAMP	NULL	Timestamp when the relationship was discharged. NULL if still

			active.
notes	TEXT	NULL	Optional clinical or administrative notes on the care relationship.
created_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Record creation timestamp.
updated_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Timestamp of last record update.

SCHEDULING 05. appointments			
Stores individual consultation session records within an RND-client relationship. Supports in-person, video, and chat modalities. Linked to invoices and NCP records.			
Column Name	Data Type	Constraints	Description
id	INT UNSIGNED	PK · AUTO_INCREMENT · NOT NULL	Surrogate primary key.
relationship_id	INT UNSIGNED	NOT NULL · FK → <i>rnd_client_relationships.id</i>	Foreign key to the governing RND-client relationship. Cascades on delete.
scheduled_at	DATETIME	NOT NULL	Scheduled date and time of the consultation.
type	ENUM	NOT NULL · DEFAULT video	Consultation modality: 'in_person', 'video', or 'chat'.
status	ENUM	NOT NULL · DEFAULT pending	Lifecycle status: 'pending', 'confirmed', 'completed', or 'cancelled'.
notes	TEXT	NULL	Optional notes about the appointment.
created_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Record creation timestamp.
updated_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Timestamp of last record update.

CLINICAL (NCP) 06. pre_consultation_screenings			
Captures client-submitted anthropometric inputs and stores the system automated computations of BMI, BMR, TDEE, and NRS-2002 nutritional risk classification prior to any RND engagement.			
Column Name	Data Type	Constraints	Description
id	INT UNSIGNED	PK · AUTO_INCREMENT · NOT NULL	Surrogate primary key.
client_id	INT UNSIGNED	NOT NULL · FK → <i>users.id</i>	Foreign key to the client who submitted the screening.

appointment_id	INT UNSIGNED	<i>NULL · FK → appointments.id</i>	Optional link to the upcoming appointment this screening relates to.
height_cm	DECIMAL(5,2)	NOT NULL	Client-entered height in centimeters.
weight_kg	DECIMAL(5,2)	NOT NULL	Client-entered weight in kilograms.
bmi	DECIMAL(5,2)	NULL	Computed Body Mass Index (weight / height ²).
bmi_category	VARCHAR(50)	NULL	WHO Asia-Pacific BMI classification (e.g., Underweight, Normal, Overweight, Obese).
bmr_kcal	DECIMAL(7,2)	NULL	Computed Basal Metabolic Rate using the Mifflin-St Jeor equation (kcal/day).
tdee_kcal	DECIMAL(7,2)	NULL	Computed Total Daily Energy Expenditure: BMR multiplied by activity factor (kcal/day).
activity_level	ENUM	NOT NULL · DEFAULT sedentary	Activity factor used for TDEE: 'sedentary', 'lightly_active', 'moderately_active', 'very_active', or 'extra_active'.
nrs_score	TINYINT(4)	NULL	Adapted NRS-2002 nutritional risk score (0–7).
nrs_risk	ENUM	NULL	Risk classification derived from NRS score: 'no_risk', 'at_risk', or 'high_risk'.
symptoms	TEXT	NULL	Client-reported symptoms entered during the screening.
created_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Timestamp when the screening was submitted.

CLINICAL (NCP) 07. ncp_records			
Encapsulates all four phases of the Nutrition Care Process per consultation encounter: Phase 1 Assessment, Phase 2 Diagnosis (PES), Phase 3 Intervention, and Phase 4 Monitoring. One record per appointment.			
Column Name	Data Type	Constraints	Description
id	INT UNSIGNED	PK · AUTO_INCREMENT · NOT NULL	Surrogate primary key.
relationship_id	INT UNSIGNED	<i>NOT NULL · FK → rnd_client_relationships.id</i>	Links this NCP record to the RND-client care relationship.

appointment_id	INT UNSIGNED	<i>NULL · FK → appointments.id</i>	Optional link to the appointment during which this record was created.
encounter_date	DATE	NOT NULL	Date of the clinical encounter.
status	ENUM	NOT NULL · DEFAULT draft	Record lifecycle: 'draft' (in progress) or 'completed' (finalized).
weight_kg	DECIMAL(5,2)	NULL	Phase 1: Measured weight in kilograms.
height_cm	DECIMAL(5,2)	NULL	Phase 1: Measured height in centimeters.
bmi	DECIMAL(5,2)	NULL	Phase 1: Computed BMI. Tracked longitudinally across records.
blood_pressure	VARCHAR(20)	NULL	Phase 1: Blood pressure reading (e.g., 120/80 mmHg).
blood_glucose	DECIMAL(5,2)	NULL	Phase 1: Fasting blood glucose in mmol/L or mg/dL.
hba1c	DECIMAL(4,2)	NULL	Phase 1: Glycated hemoglobin percentage.
lab_notes	TEXT	NULL	Phase 1: Free-text notes on laboratory results.
assessment_notes	TEXT	NULL	Phase 1: Clinical assessment narrative.
pes_problem	VARCHAR(500)	NULL	Phase 2: Problem component of the PES statement (standardized nutrition diagnosis term).
pes_etiology	TEXT	NULL	Phase 2: Etiology — contributing factors to the nutrition problem.
pes_signs	TEXT	NULL	Phase 2: Signs and Symptoms — measurable evidence supporting the diagnosis.
diet_prescription	TEXT	NULL	Phase 3: Written diet prescription (e.g., 1800 kcal diabetic diet).
target_kcal	DECIMAL(7,2)	NULL	Phase 3: Target daily energy intake in kilocalories.
target_protein_g	DECIMAL(6,2)	NULL	Phase 3: Target daily protein intake in grams.
target_carb_g	DECIMAL(6,2)	NULL	Phase 3: Target daily carbohydrate intake in

			grams.
target_fat_g	DECIMAL(6,2)	NULL	Phase 3: Target daily fat intake in grams.
intervention_notes	TEXT	NULL	Phase 3: Additional intervention guidance or care plan notes.
monitoring_notes	TEXT	NULL	Phase 4: Monitoring observations and outcome notes.
goal_status	ENUM	NULL	Phase 4: Goal achievement status: 'met', 'partially_met', 'not_met', or 'ongoing'.
created_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Record creation timestamp.
updated_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Timestamp of last record update.

CLINICAL (NCP) 08. meal_plans			
RND-created meal plan prescriptions tied to a specific care relationship. Defines daily exchange allowances per food group for a targeted clinical condition. Distributed across six meal times via meal_plan_meals.			
Column Name	Data Type	Constraints	Description
id	INT UNSIGNED	PK · AUTO_INCREMENT · NOT NULL	Surrogate primary key.
relationship_id	INT UNSIGNED	NOT NULL · FK → <i>rnd_client_relationships.id</i>	Foreign key to the RND-client relationship this meal plan is prescribed for.
name	VARCHAR(255)	NOT NULL	Descriptive name of the meal plan (e.g., Week 1 Diabetic Plan - 1800 kcal).
condition	ENUM	NOT NULL · DEFAULT general	Target clinical condition: 'diabetes', 'hypertension', 'renal', 'weight_loss', 'weight_gain', or 'general'.
target_kcal	DECIMAL(7,2)	NULL	Target total daily energy intake in kilocalories.
total_vegetable	DECIMAL(4,1)	NULL · DEFAULT 0.0	Total daily vegetable exchanges prescribed across all meal times.
total_fruit	DECIMAL(4,1)	NULL · DEFAULT 0.0	Total daily fruit exchanges prescribed.
total_milk	DECIMAL(4,1)	NULL · DEFAULT 0.0	Total daily milk exchanges prescribed.
total_rice	DECIMAL(4,1)	NULL · DEFAULT 0.0	Total daily rice/starch

			exchanges prescribed.
total_meat	DECIMAL(4,1)	NULL · DEFAULT 0.0	Total daily meat/protein exchanges prescribed.
total_fat	DECIMAL(4,1)	NULL · DEFAULT 0.0	Total daily fat exchanges prescribed.
total_sugar	DECIMAL(4,1)	NULL · DEFAULT 0.0	Total daily sugar exchanges prescribed.
notes	TEXT	NULL	Optional clinical notes on the overall meal plan.
status	ENUM	NOT NULL · DEFAULT active	Plan lifecycle: 'active' or 'archived'.
created_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Record creation timestamp.
updated_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Timestamp of last record update.

CLINICAL (NCP) 09. meal_plan_meals			
Distributes total exchange allowances from a meal plan across six meal times (Breakfast, AM Snack, Lunch, PM Snack, Dinner, Bedtime Snack), following the FNRI Food Exchange List meal planning form layout.			
Column Name	Data Type	Constraints	Description
id	INT UNSIGNED	PK · AUTO_INCREMENT · NOT NULL	Surrogate primary key.
meal_plan_id	INT UNSIGNED	NOT NULL · FK → meal_plans.id	Foreign key to the parent meal plan. Cascades on delete.
meal_time	ENUM	NOT NULL	Meal time slot: 'breakfast', 'am_snack', 'lunch', 'pm_snack', 'dinner', or 'bedtime_snack'. Pair (meal_plan_id + meal_time) is UNIQUE.
vegetable_exchanges	DECIMAL(4,1)	NOT NULL · DEFAULT 0.0	Vegetable exchanges allocated to this meal time.
fruit_exchanges	DECIMAL(4,1)	NOT NULL · DEFAULT 0.0	Fruit exchanges allocated to this meal time.
milk_exchanges	DECIMAL(4,1)	NOT NULL · DEFAULT 0.0	Milk exchanges allocated to this meal time.
rice_exchanges	DECIMAL(4,1)	NOT NULL · DEFAULT 0.0	Rice/starch exchanges allocated to this meal time.

meat_exchanges	DECIMAL(4,1)	NOT NULL · DEFAULT 0.0	Meat/protein exchanges allocated to this meal time.
fat_exchanges	DECIMAL(4,1)	NOT NULL · DEFAULT 0.0	Fat exchanges allocated to this meal time.
sugar_exchanges	DECIMAL(4,1)	NOT NULL · DEFAULT 0.0	Sugar exchanges allocated to this meal time.
meal_notes	TEXT	NULL	Optional notes for this meal time (e.g., Take with medication).

CLINICAL (NCP)			
10. meal_plan_food_items			
Specific food items selected for each meal time slot. References food_exchange_items for FNRI-listed foods. The food_name field captures items from the FNRI FCT API or USDA FoodData Central API — only the name is persisted; nutrient data remains transient.			
Column Name	Data Type	Constraints	Description
id	INT UNSIGNED	PK · AUTO_INCREMENT · NOT NULL	Surrogate primary key.
meal_plan_meal_id	INT UNSIGNED	NOT NULL · FK → meal_plan_meals.id	Foreign key to the meal time slot this food item belongs to. Cascades on delete.
food_item_id	INT UNSIGNED	NULL · FK → food_exchange_items.id	Optional FK to a locally stored FNRI food item. NULL for items sourced from external APIs.
food_name	VARCHAR(255)	NOT NULL	Display name of the food item. Populated from the local table for FNRI items, or from the FNRI FCT / USDA API for external items. Only persisted fields from external APIs.
exchanges	DECIMAL(4,1)	NOT NULL · DEFAULT 1.0	Number of exchange units of this food item assigned to the meal time.
household_measure	VARCHAR(100)	NULL	Household measurement equivalent (e.g., 1/2 cup cooked) for client-facing meal plans.
notes	VARCHAR(500)	NULL	Preparation or serving instructions specific to this food item in the meal plan.

CLINICAL (NCP)			
11. progress_records			
Longitudinal tracking of key clinical and behavioral outcome indicators between or after consultations. Feeds the progress monitoring dashboard and automated outcome reports visible to both RND and client.			
Column Name	Data Type	Constraints	Description
id	INT UNSIGNED	PK · AUTO_INCREMENT · NOT NULL	Surrogate primary key.
relationship_id	INT UNSIGNED	NOT NULL · FK → <i>rnd_client_relationships.id</i>	Foreign key to the RND-client relationship this progress record belongs to.
record_date	DATE	NOT NULL	Date the progress metrics were logged.
weight_kg	DECIMAL(5,2)	NULL	Logged body weight in kilograms.
blood_pressure	VARCHAR(20)	NULL	Blood pressure reading (e.g., 118/76 mmHg).
blood_glucose	DECIMAL(5,2)	NULL	Blood glucose value in mmol/L or mg/dL.
hba1c	DECIMAL(4,2)	NULL	Glycated hemoglobin percentage.
adherence_pct	TINYINT UNSIGNED	NULL	Estimated dietary adherence percentage (0–100) for the period.
client_notes	TEXT	NULL	Client-submitted notes or self-report for this record.
rnd_notes	TEXT	NULL	RND clinical observations or notes for this record.
created_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Timestamp when the progress record was created.

COMMUNICATION & RESOURCES			
12. messages			
RA 10173-compliant in-platform clinical messaging scoped to each RND-client relationship. Supports secure, auditable communication between practitioners and patients.			
Column Name	Data Type	Constraints	Description
id	INT UNSIGNED	PK · AUTO_INCREMENT · NOT NULL	Surrogate primary key.
relationship_id	INT UNSIGNED	NOT NULL · FK → <i>rnd_client_relationships.id</i>	Foreign key scoping the message to a specific RND-client relationship.
sender_id	INT UNSIGNED	NOT NULL · FK → <i>users.id</i>	Foreign key identifying which user (RND or client) sent the

			message.
message	TEXT	NOT NULL	Message body content.
is_read	TINYINT(1)	NOT NULL · DEFAULT 0	Read receipt flag. 0 = unread; 1 = read by the recipient.
created_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Timestamp of message creation. Used to sort messages chronologically.

COMMUNICATION & RESOURCES			
13. resources			
Nutrition education materials and physical activity resources uploaded by RNDs or administrators. Supports the digital resource repository accessible to matched clients.			
Column Name	Data Type	Constraints	Description
id	INT UNSIGNED	PK · AUTO_INCREMENT · NOT NULL	Surrogate primary key.
uploaded_by	INT UNSIGNED	NOT NULL · FK → users.id	Foreign key to the user (RND or admin) who uploaded or registered this resource.
title	VARCHAR(500)	NOT NULL	Descriptive title of the resource.
description	TEXT	NULL	Optional summary or description of the resource content.
type	ENUM	NOT NULL · DEFAULT article	Resource format: 'article', 'pdf', 'video', or 'link'.
file_path	VARCHAR(500)	NULL	Server file path for uploaded files. NULL for link-type resources.
url	VARCHAR(1000)	NULL	External URL for link-type and video-type resources. NULL for uploaded files.
is_active	TINYINT(1)	NOT NULL · DEFAULT 1	Visibility flag. 0 = hidden; 1 = accessible to clients.
created_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Timestamp when the resource was uploaded.

COMMUNICATION & RESOURCES			
14. reminders			
Scheduled notification records sent to clients to support behavioral adherence between consultations. Covers appointment reminders, meal logging prompts, medication cues, and general habit-support messages.			
Column Name	Data Type	Constraints	Description

id	INT UNSIGNED	PK · AUTO_INCREMENT · NOT NULL	Surrogate primary key.
client_id	INT UNSIGNED	<i>NOT NULL · FK → users.id</i>	Foreign key to the client recipient. Cascades on delete.
title	VARCHAR(255)	NOT NULL	Short title of the reminder shown in the notification.
message	TEXT	NULL	Full message body of the reminder notification.
type	ENUM	NOT NULL · DEFAULT general	Reminder category: 'appointment', 'meal_log', 'medication', or 'general'.
send_at	DATETIME	NOT NULL	Scheduled delivery datetime. Processed by the Laravel scheduler at the specified time.
is_sent	TINYINT(1)	NOT NULL · DEFAULT 0	Delivery status flag. 0 = pending; 1 = dispatched by the notification service.
created_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Timestamp when the reminder was scheduled.

BILLING & FINANCIAL			
15. invoices			
Tracks consultation fees, platform commission percentages and computed amounts, and payment status. Linked to both the RND-client relationship and optionally to a specific appointment. Commission amount is frozen at invoice creation to prevent retroactive adjustment.			
Column Name	Data Type	Constraints	Description
id	INT UNSIGNED	PK · AUTO_INCREMENT · NOT NULL	Surrogate primary key.
relationship_id	INT UNSIGNED	<i>NOT NULL · FK → rnd_client_relationships.id</i>	Foreign key to the RND-client relationship this invoice belongs to.
appointment_id	INT UNSIGNED	<i>NULL · FK → appointments.id</i>	Optional link to the specific appointment being invoiced.
amount	DECIMAL(10,2)	NOT NULL	Gross consultation fee charged to the client (PHP).
commission_pct	DECIMAL(5,2)	NOT NULL · DEFAULT 10.00	Platform commission percentage at time of invoice creation. Frozen to prevent retroactive changes.
commission_amt	DECIMAL(10,2)	NOT NULL · DEFAULT 0.00	Computed commission amount (amount × commission_pct / 100). Frozen at creation.

status	ENUM	NOT NULL · DEFAULT unpaid	Payment status: 'unpaid', 'paid', or 'cancelled'.
paid_at	TIMESTAMP	NULL	Timestamp of successful payment. NULL if unpaid or cancelled.
notes	TEXT	NULL	Optional notes on the invoice or payment.
created_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Timestamp when the invoice was generated.

BILLING & FINANCIAL			
16. reviews			
Records client-submitted ratings and written feedback on the RND, constrained to one review per care relationship. Ratings appear on the RND public profile to support informed client decision-making in the matching module.			
Column Name	Data Type	Constraints	Description
id	INT UNSIGNED	PK · AUTO_INCREMENT · NOT NULL	Surrogate primary key.
relationship_id	INT UNSIGNED	NOT NULL · UK · FK → <i>rnd_client_relationships.id</i>	Foreign key to the care relationship. UNIQUE constraint ensures one review per relationship.
rating	TINYINT UNSIGNED	NOT NULL	Numeric rating submitted by the client (1–5 stars).
comment	TEXT	NULL	Optional written feedback accompanying the rating.
created_at	TIMESTAMP	NOT NULL · DEFAULT NOW()	Timestamp when the review was submitted.

COMPLIANCE & ADMIN			
17. audit_logs			
Immutable system-generated trail of all significant user and administrative actions across every module. Fulfills the audit logging mandate of RA 10173 and the NPC compliance advisory for digital health platforms. Rows are never updated or deleted.			
Column Name	Data Type	Constraints	Description
id	INT UNSIGNED	PK · AUTO_INCREMENT · NOT NULL	Surrogate primary key.
user_id	INT UNSIGNED	NULL · FK → <i>users.id</i>	Foreign key to the acting user. NULL for system-generated events (e.g., automated reminders).
action	VARCHAR(100)	NOT NULL	Short machine-readable action identifier (e.g., user.login, ncp.finalized, invoice.paid).

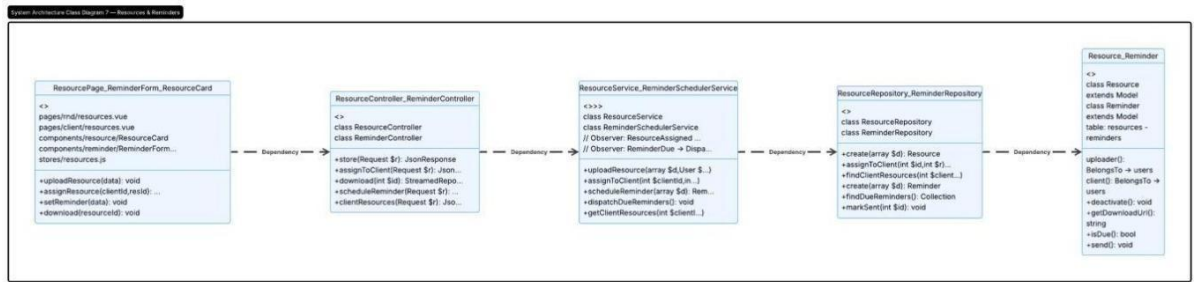


Figure 40. Resources and Reminders

UI/UX

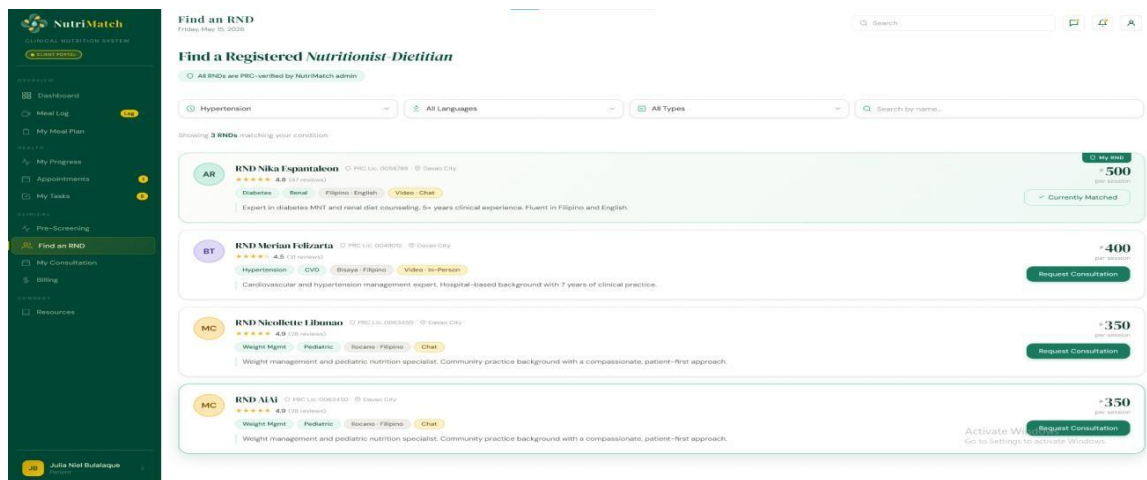


Figure 41. RND-Client Matching

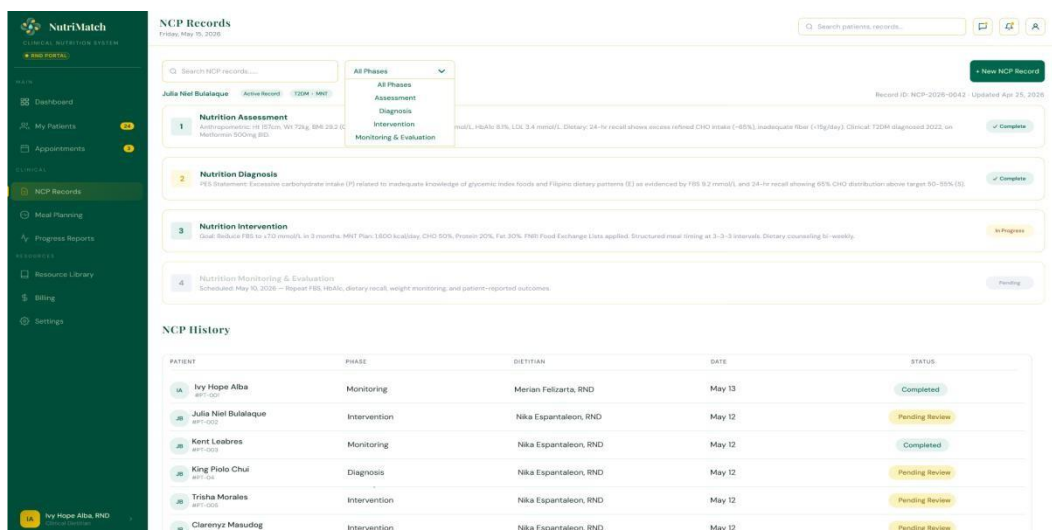


Figure 42. RND NCP Records

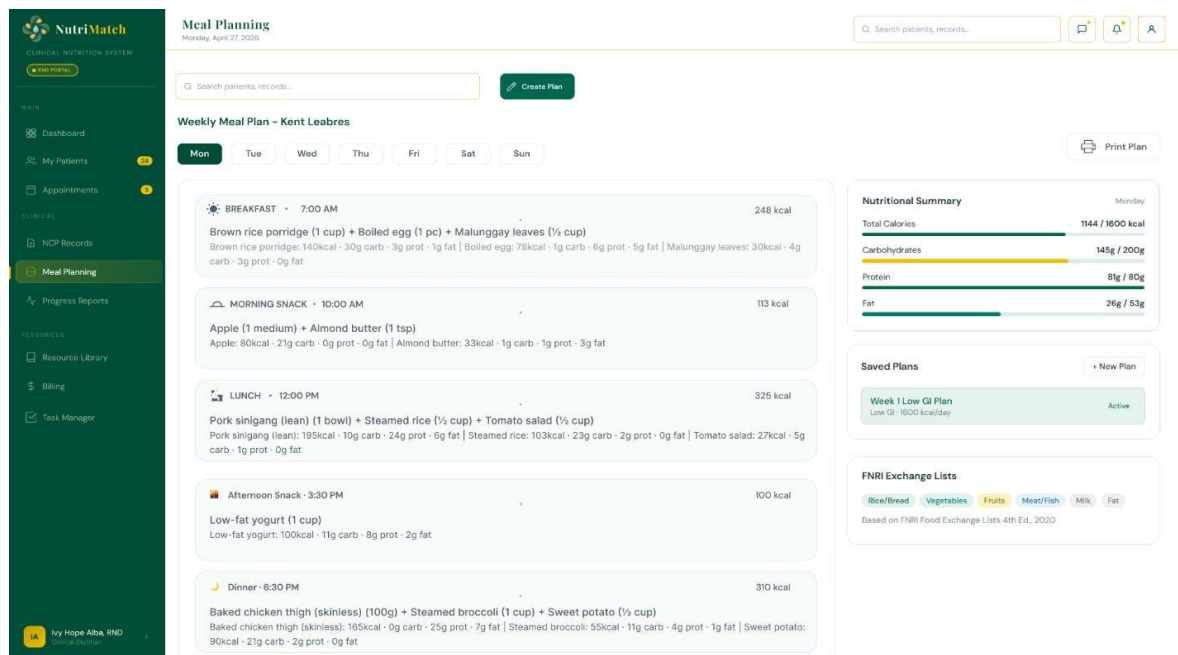


Figure 43. RND Meal-Planning

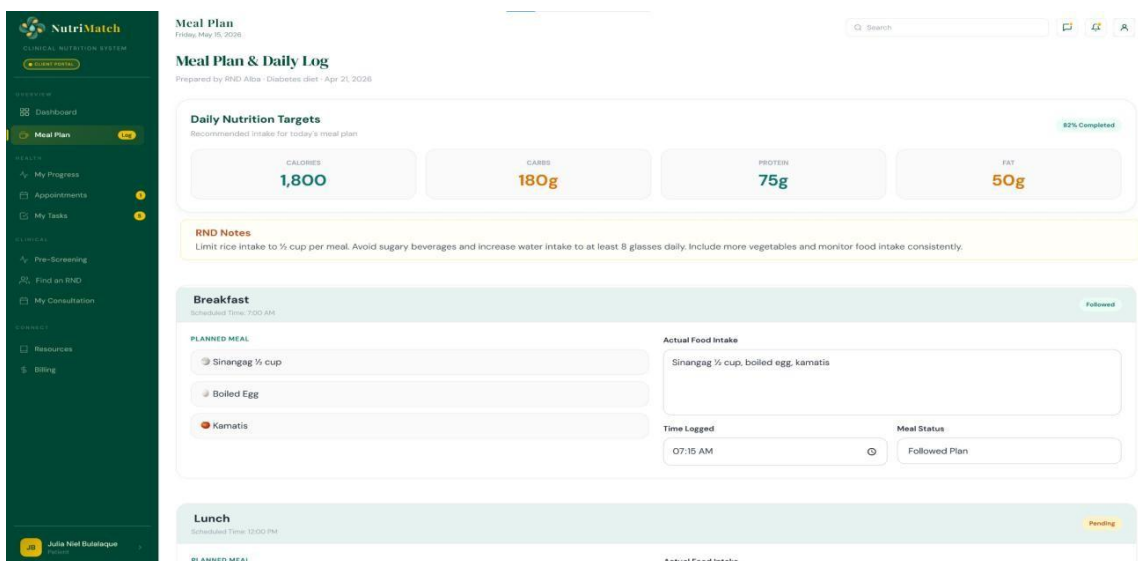


Figure 44. Client Meal Plan

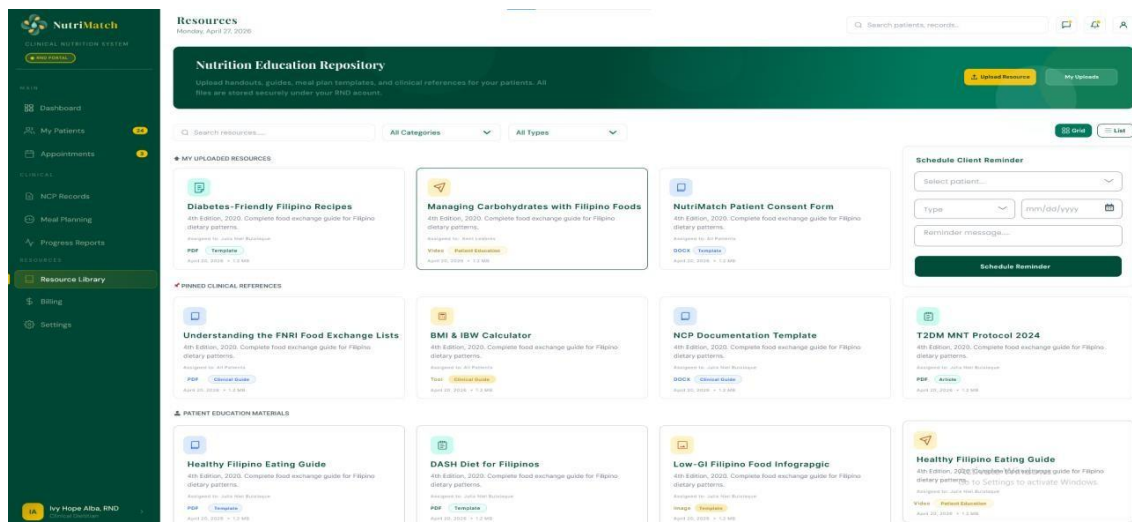


Figure 45. RND Resource and Reminder

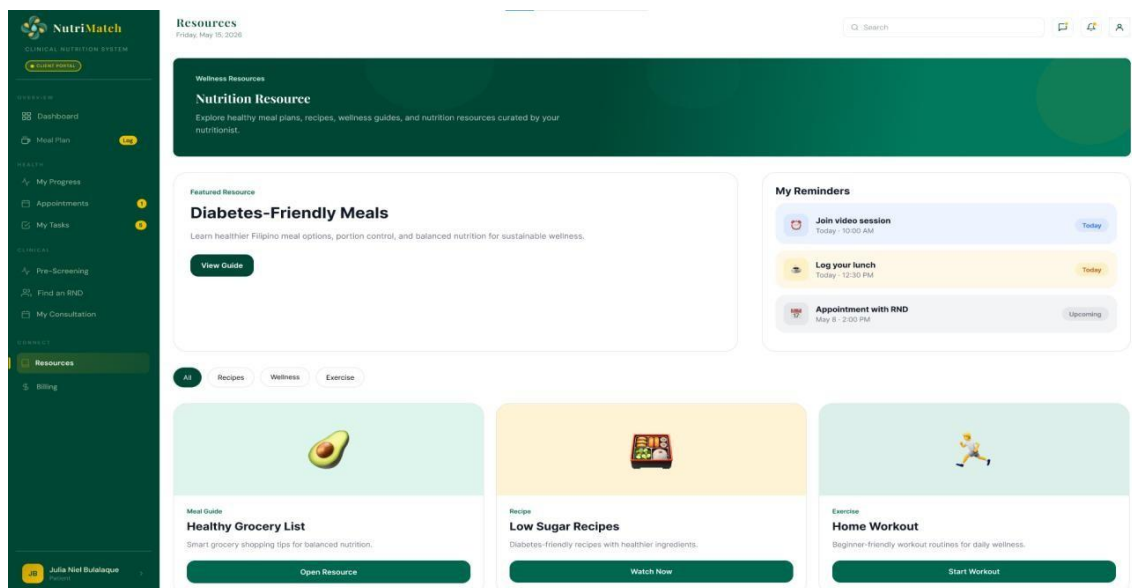


Figure 46. Client Resource and Reminder

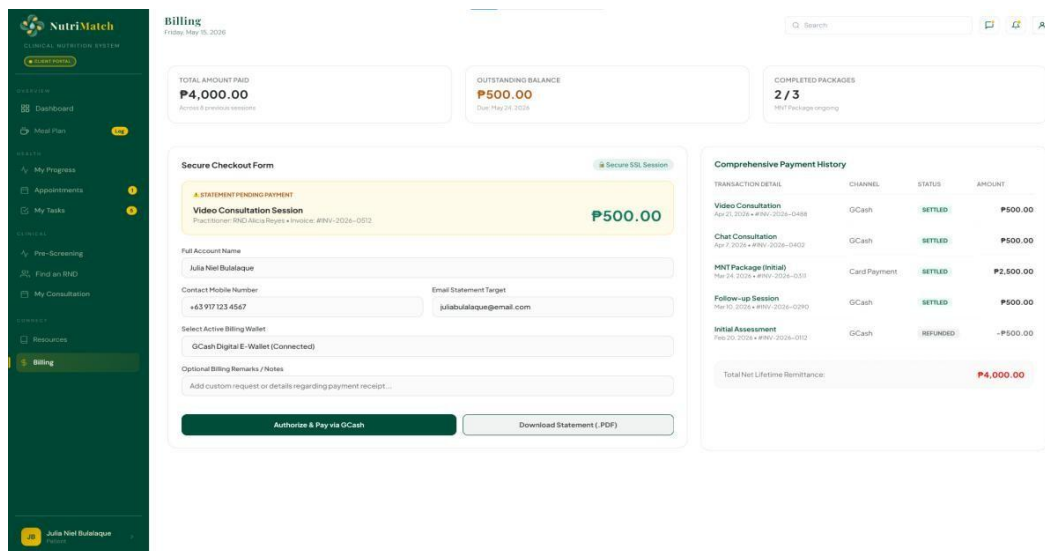


Figure 47. Client Billing

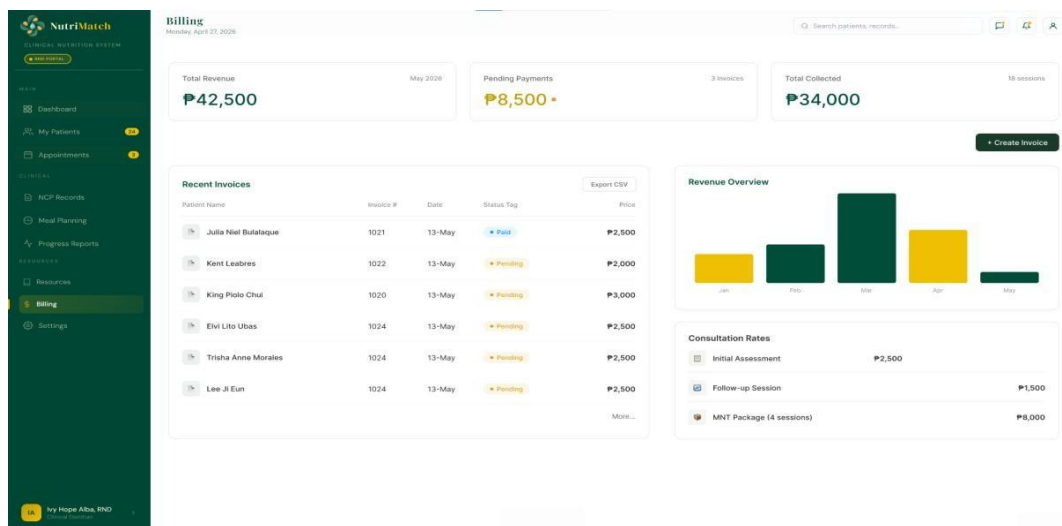


Figure 48. RND Billing

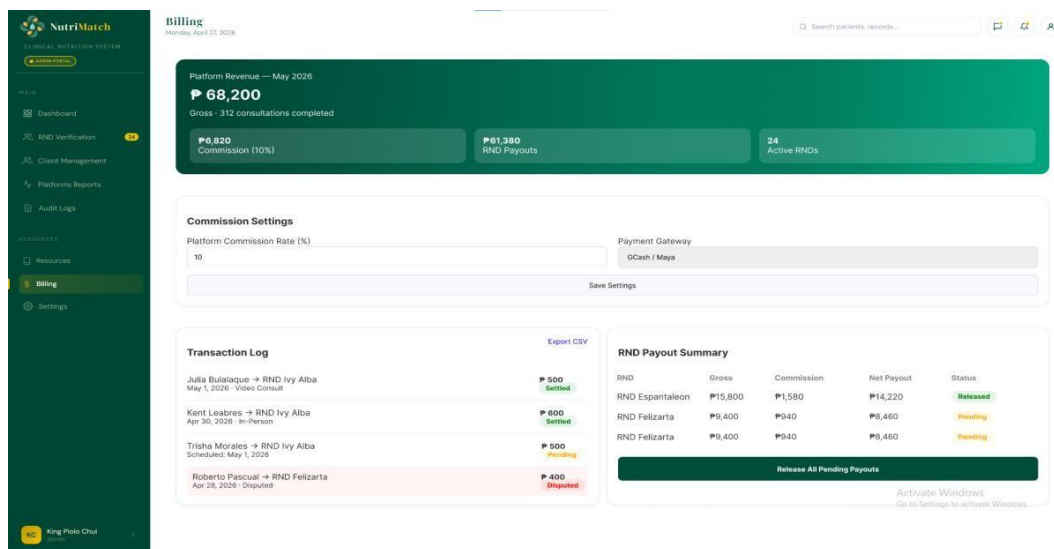


Figure 49. Admin Billing and Commission

APPENDIX A. RESOURCE PERSONS

Francis Jade M. Solomon

Capstone Project Coordinator

Elvi Lito Ubas

Capstone Project Adviser

Prince Immanuel L. Chiu

Registered Nutritionist-Dietitian

Joerine Trexie Jimenez

Registered Nutritionist-Dietitian

Gyle Francis Forones

Registered Nutritionist-Dietitian

APPENDIX B. PERSONAL TECHNICAL VITAE

Curriculum Vitae of
King Paolo L. Chiu
 44 Mercury Street GSIS Heights Matina, Davao City
 chiu.429953@davao.sti.edu.ph
 09177270495

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023-present	STI College of Davao
Vocational/Technical	N/A	N/A
High School	2007-2011	Daniel R. Aguinaldo National High School
Elementary	2002-2007	Davao City Special School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
12/2018-01/2023	Back Office Support	FGC +; 11 Bonifacio St, Poblacion District, Davao City, 8000 Davao del Sur
06/2018-12/2018	Account Supervisor	FGC +; 11 Bonifacio St, Poblacion District, Davao City, 8000 Davao del Sur
02/2018-06/2018	Account Specialist	FocusInc Group; 11 Bonifacio St, Poblacion District, Davao City, 8000 Davao del Sur
08/2016-02/2018	Records Retrieval VA	FocusInc Group; 11 Bonifacio St, Poblacion District, Davao City, 8000 Davao del Sur

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2026	OSOA	Member

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired month year month year month year
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TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
09/2023	The Modern Stack How do you succeed as a Programmer?
09/2023	Unveiling the AI Advantage for Next-Gen Contact Center
10/2023	What is it like to work in cybersecurity?
10/2023	Patch-Based Image Stitching for UAV Images with Natural Scene Distortions

Listed in reverse chronological order (most recent first).

Curriculum Vitae of
JULIA NIEL BULALAUQUE
Purok 22, Brgy 76-A, Santan St. Ecoland, Davao City
bulalauque.334427@davao.sti.edu.ph
09663394751

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023-present	STI College of Davao
Vocational/Technical	2021-2023	St. John Paul II College of Davao
High School	2017-2021	St. Ana National High School
Elementary	2011-2017	St. Ana Central Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
07/202 –02/2023	General Assistant	PRL BPO Company

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2026	OSOA	Member

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired month year month year month year
--------	---------------------	---

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
month year	
month year	
month year	
month year	

Listed in reverse chronological order (most recent first).

Curriculum Vitae of
IVY HOPE O. ALBA
 Garcia Heights, Bacaca Road, Bajada, Davao City
 alba.342335@davao.sti.edu.ph
 09051372423

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023-present	STI College of Davao
Vocational/Technical	2020-2022	Crispin E. Rojas National High School
High School	2016-2020	Crispin E. Rojas National High School
Elementary	2010-2016	Lambajon Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
01/2022-06/2023	Sales Agent	ACME

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2026	OSOA	Member

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired month year month year month year
--------	---------------------	---

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
month year	
month year	
month year	
month year	

Listed in reverse chronological order (most recent first).

Curriculum Vitae of
KENT DARYLL V. LEABRES
 Juan Luna Street Poblacion Dos, Bansalan
 leabres.371033@davao.sti.edu.ph
 09617449347

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023	STI College of Davao
Vocational/Technical	N/A	N/A
High School	2017-2020	Nazarenus College and Foundation Incorporation Marilao City
Elementary	2010-2016	Calvario Elementary School Meycauayan City

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
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Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates month year	Name of Organization	Position
-------------------------------	----------------------	----------

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired month year month year month year
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TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates month year month year month year month year	Title of Training, Seminar, or Workshop
---	---

Listed in reverse chronological order (most recent first).